

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD		√	√
V810 / V810i XXL		√	√
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 Apr 2024	First Release	Yeap Leong Wei
01	30 Oct 2024	Include STD & XXL for applicable model + content update	Yeap Leong Wei

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

Content

Content.....	2
Scope	3
Tool List	3
For LE, AKD & S200 Motion Drive Only	4
X-Axis Home and Limit Sensor Alignment for LE, AKD and S200 Drive Only	5
Y Axis Home and Limit Sensor Alignment for LE, AKD and S200 Drive Only	11
For Panasonic Motion Drive Only	17
X-Axis Home and Limit Sensor Alignment for Panasonic Drive	18
Y-Axis Home and Limit Sensor Alignment for Panasonic Drive	31
Validation of Machine Performance After Alignment for Sensors	44

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

Scope

This procedure is designed to provide guidelines for the alignment of home and limit sensors paired with the Quantic Scale.

Tool List

No	PN	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Camera Tool	1	PC	

Table 1: Tool List

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

For LE, AKD & S200 Motion Drive Only

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

X-Axis Home and Limit Sensor Alignment for LE, AKD and S200 Drive Only

Note: for Panasonic Drive, proceed to [link](#)

1. **Launch the v810:** Find the v810 shortcut icon at the desktop window and launch the program.



Figure 1: v810 shortcut

2. **Digital IO Initialization:** Navigate to ‘Service > Subsystem Status & Control > Digital IO tab’. Click “Initialize” and wait for the subsystem initialization to complete.

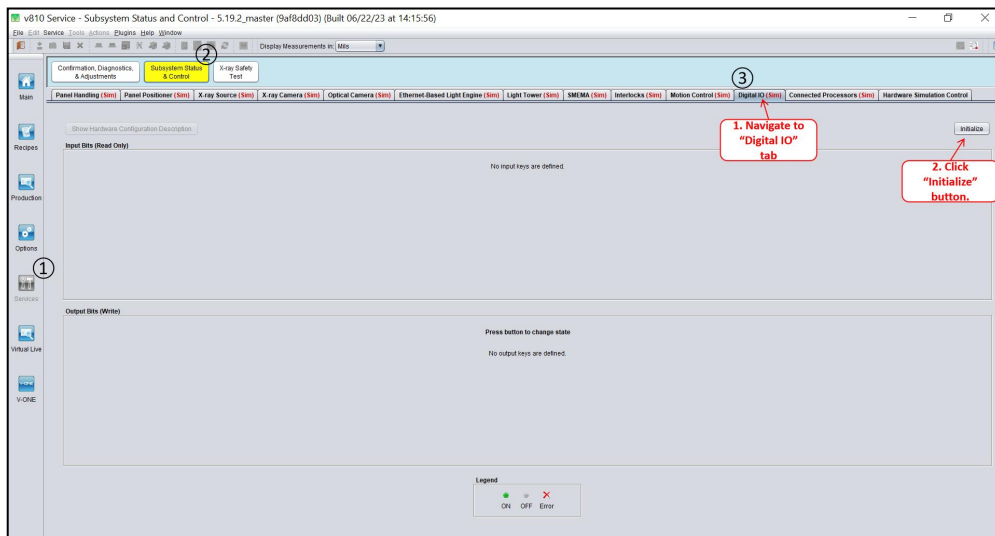


Figure 2: DIO Initialization (Sample Image)

3. **Panel Positioner Initialization:** Upon completion of DIO initialize, navigate to ‘Panel Positioner’ tab and click “Initialize”.

#IMPORTANT This step must be successfully executed before proceed to sensors adjustment.

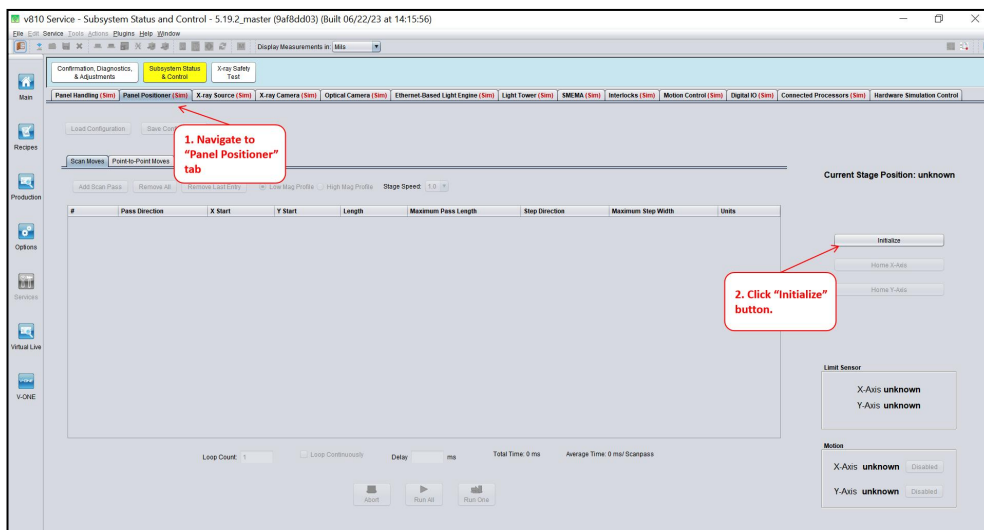


Figure 3: Initialize “Panel Positioner” (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

4. **Verify the “Current Stage Position” for X-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the X stage to the hard stop (maximum left) position and check if X-axis is within 3.08mm to 3.28mm. If not within the range, proceed to Step 5. Else, proceed to Step 8.

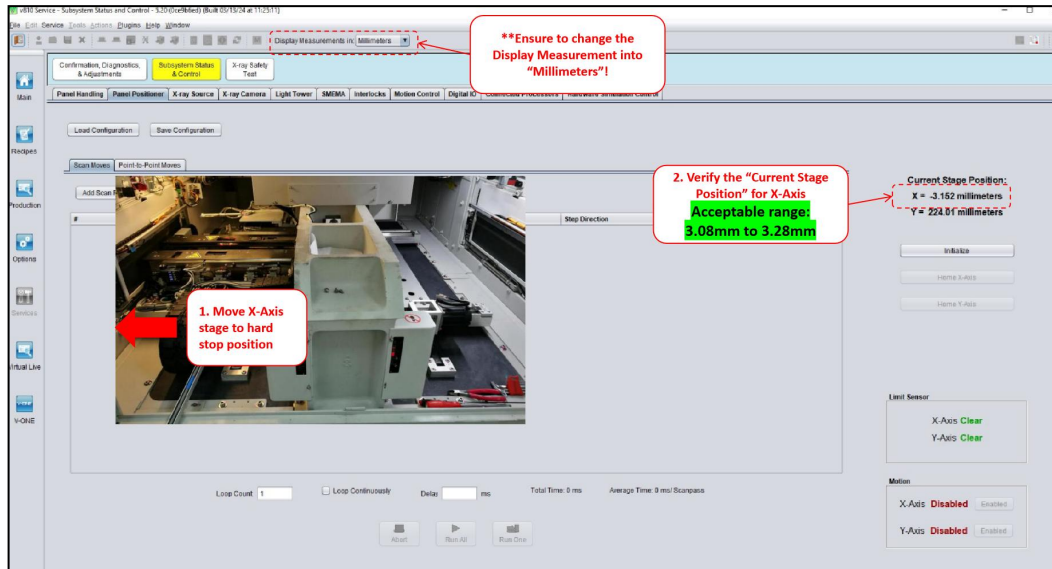


Figure 4: Current Stage Position for X-Axis within range (Sample Image)

5. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp on X-axis.

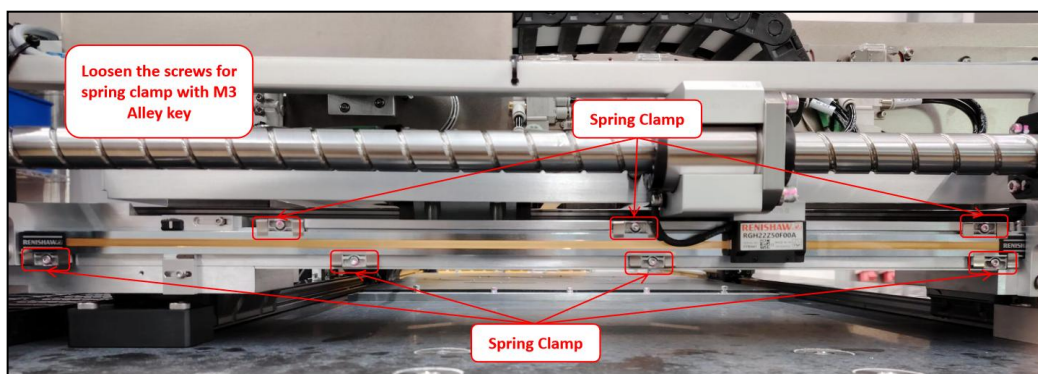


Figure 5: Quantic Scale Spring Clamps (Sample Image)

6. **Glass Scale Positioning Adjustment:** Gently move the glass scale following the direction indicated in the table below until the x-axis current position reading in V810 fall in the range between 3.08mm to 3.28mm.

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.08mm	Towards the right outer barrier direction
More than -3.28mm	Towards the left outer barrier direction

Table 2: X-Axis Glass Scale Positioning Adjustment Guidelines

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

7. **Tighten the Spring Clamps:** Upon completion of X-Axis glass scale alignment, tighten all screw for the spring clamps.

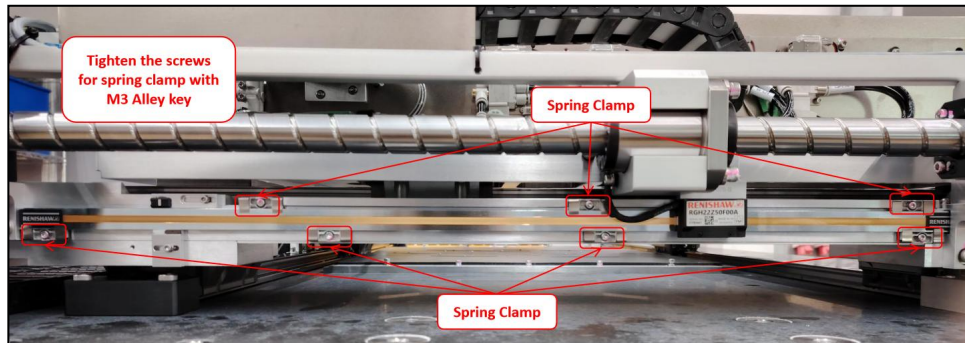


Figure 6: X-Axis glass scale Spring clamp (Sample Image)

8. **Home Sensor Positioning Offset:** Loosen the screw on the X-axis home sensor. Position the left side of the home sensor on the engraved line marking at the top side of the glass scale. Tighten back the screw on the X-axis home sensor when done.

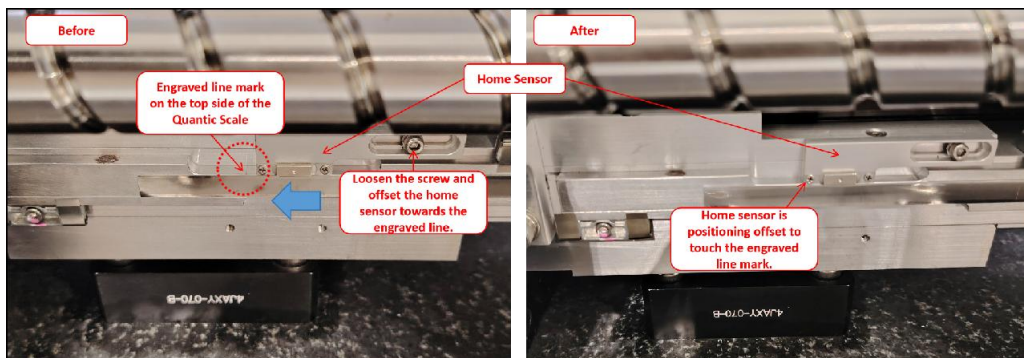


Figure 7: X-Axis Home Sensor Offset

9. **X-Axis Positioning (Away from Hard Stop):** Move the X-axis to the middle of the stage.

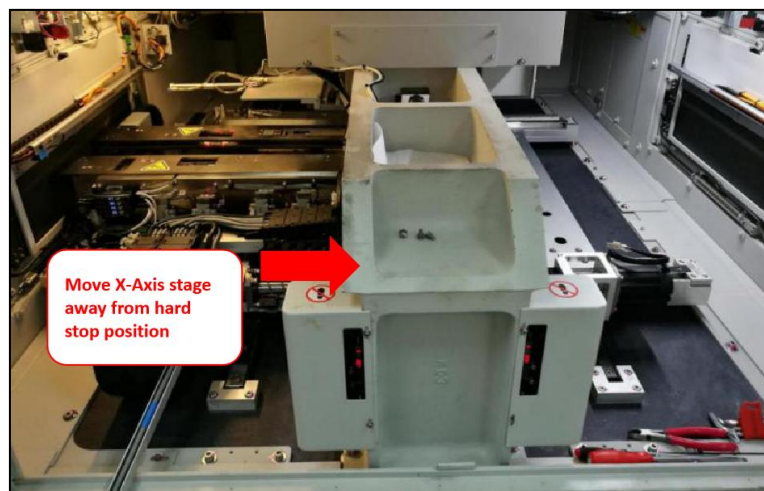


Figure 8: Move X-Axis Stage Away From Hard Stop

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors	Effective Date	30 Oct 2024
	Alignment Procedure		

10. **X-Axis Homing:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for X-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home X-Axis”

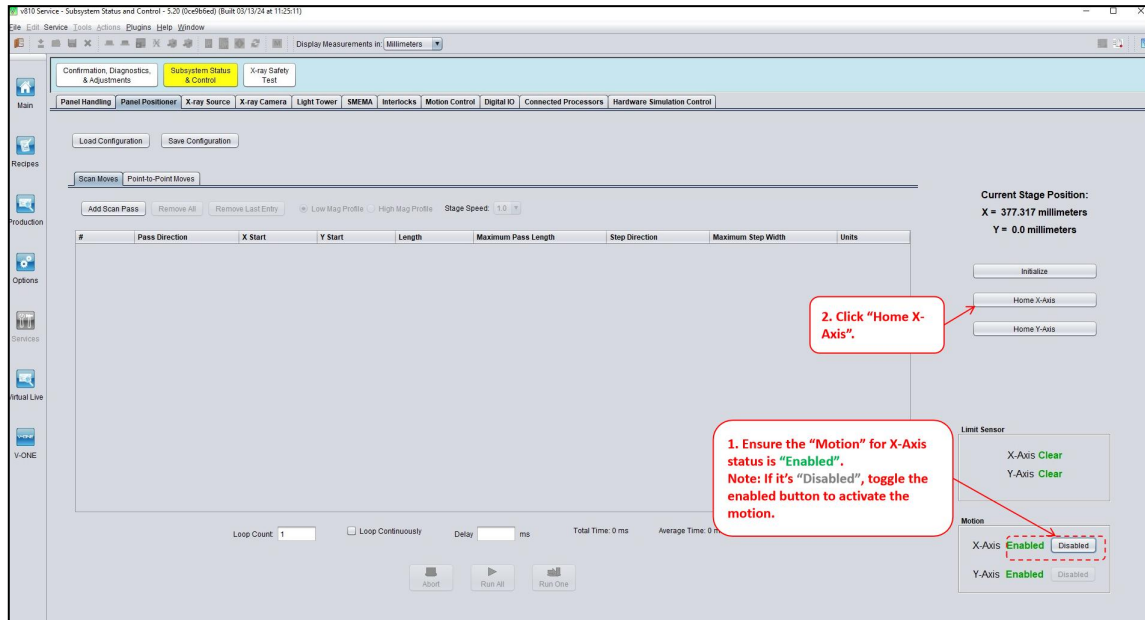


Figure 9: X-Axis Homing

11. **X-Axis “Current Stage Position” Verification (Hard Stop Position):** Move the X-Axis stage to hard stop position, double confirm the current stage position is within the range of 3.08mm to 3.28mm. If current position of X-Axis is out of range from 3.08mm to 3.28mm, move the x-stage to center and perform x-axis homing. Then, route back to Step 5. Else, proceed to Step 12.

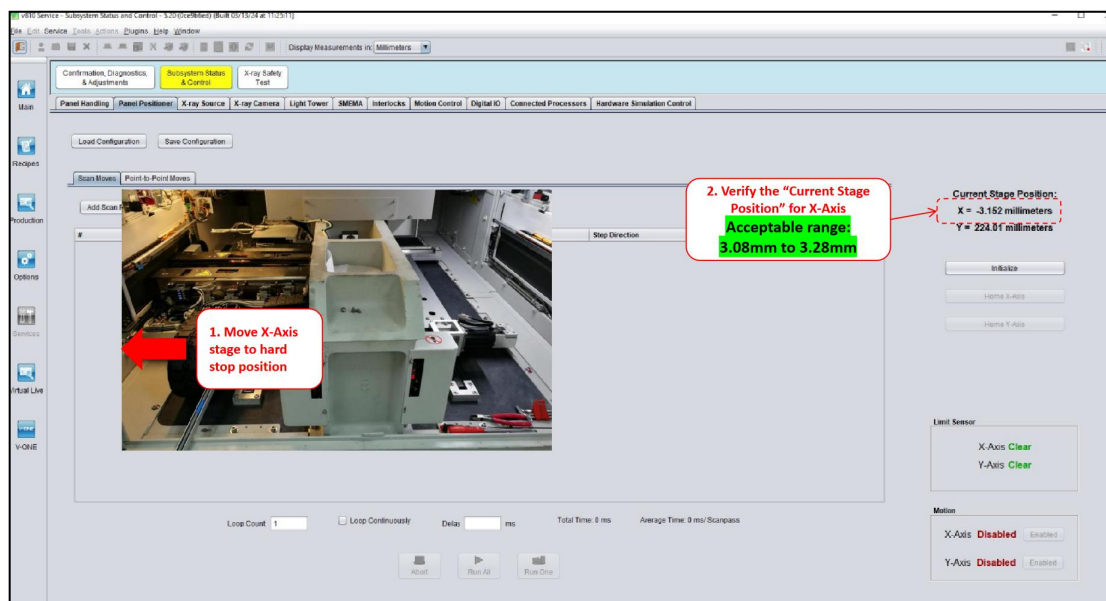


Figure 10: Verification of Current Stage Position for X-Axis

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors	Effective Date	30 Oct 2024
	Alignment Procedure		

12. **Homing again:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for X-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home X-Axis”

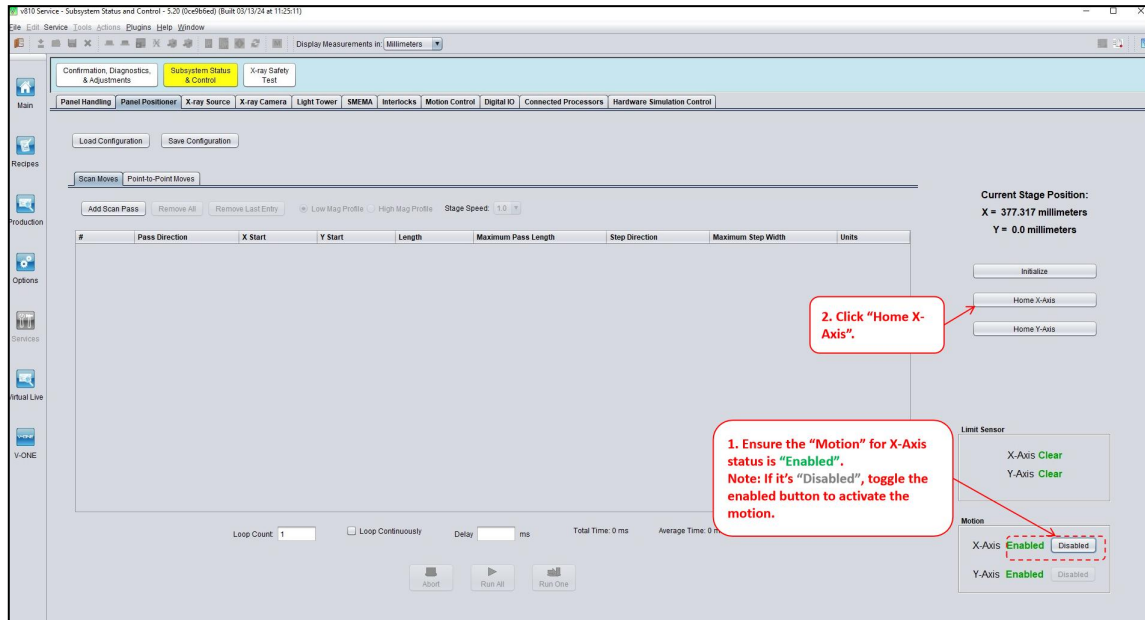


Figure 11: Perform Homing

13. **Move the Stage towards Hard Stop direction until it hit -0.5mm:** Move the stage towards the hard stop direction and ensure the current stage position is stop at -0.5mm.

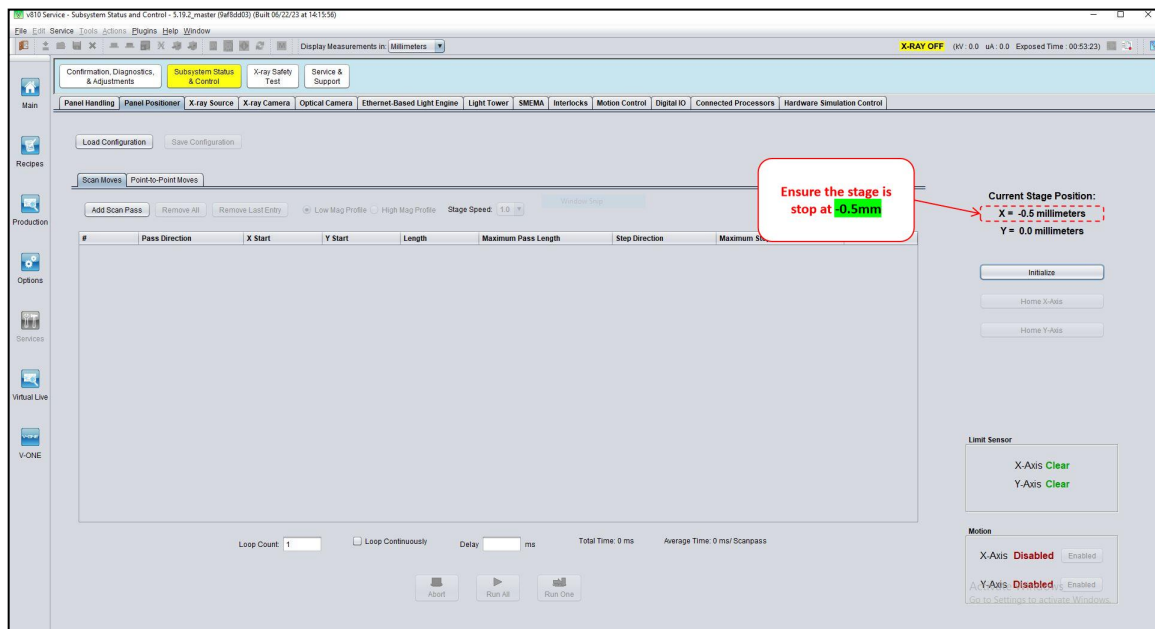


Figure 12: Move X-Axis Stage to Current Stage Position -0.5mm (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

14. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

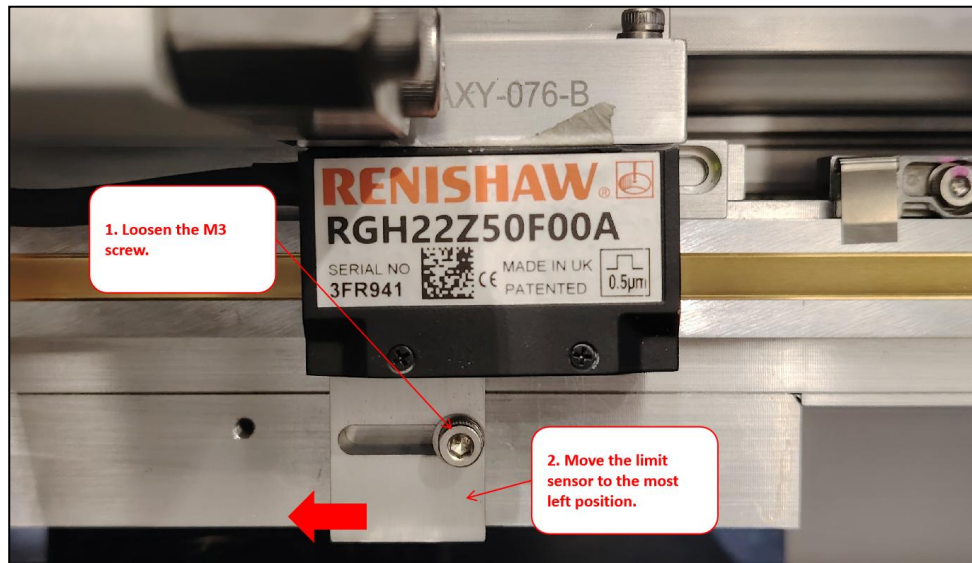


Figure 13: Move Limit Sensor to the Most Left Position

15. **Adjust the limit sensor to right hand side until the signal is “Blocked”:** Slowly move the limit sensor towards the right hand side position until the signal for limit sensor is blocked.

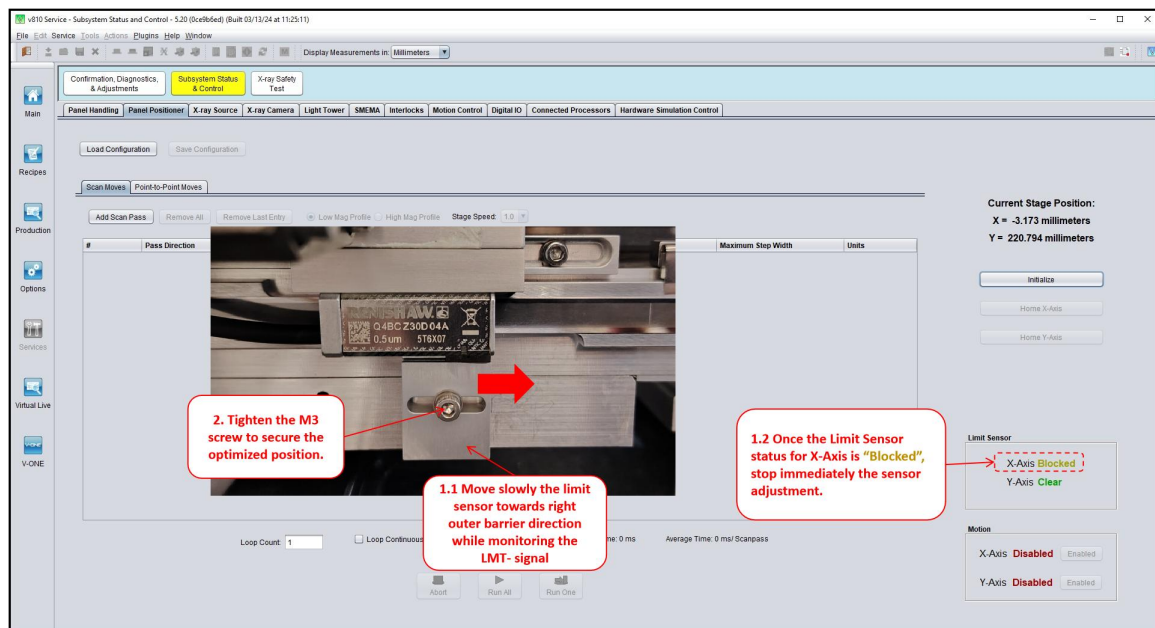


Figure 14: Positioning Limit Sensor to Triggered State

16. **Final Round Verification:** Lastly, manually move the X-stage to the center position, perform homing and ensure the signal for limit sensor is “Clear”. Then, manually move the stage to hard stop position and ensure the signal for limit sensor is “Blocked”.
17. **Validation of Machine Performance After Adjustment and Confirmation:** Machine performance validation is required before release back to customer.

Note: If there is involvement of Y-axis scale replacement, validation may be done upon all adjustments made.

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

Y Axis Home and Limit Sensor Alignment for LE, AKD and S200 Drive Only

Note: for Panasonic Drive, proceed to [link](#)

18. **Launch the V810:** Find the v810 shortcut icon at the desktop window and launch the program.



Figure 15: v810 shortcut

19. **Digital IO Initialization:** Navigate to ‘Service > Subsystem Status & Control > Digital IO tab’. Click “Initialize” and wait for the subsystem initialization to complete.

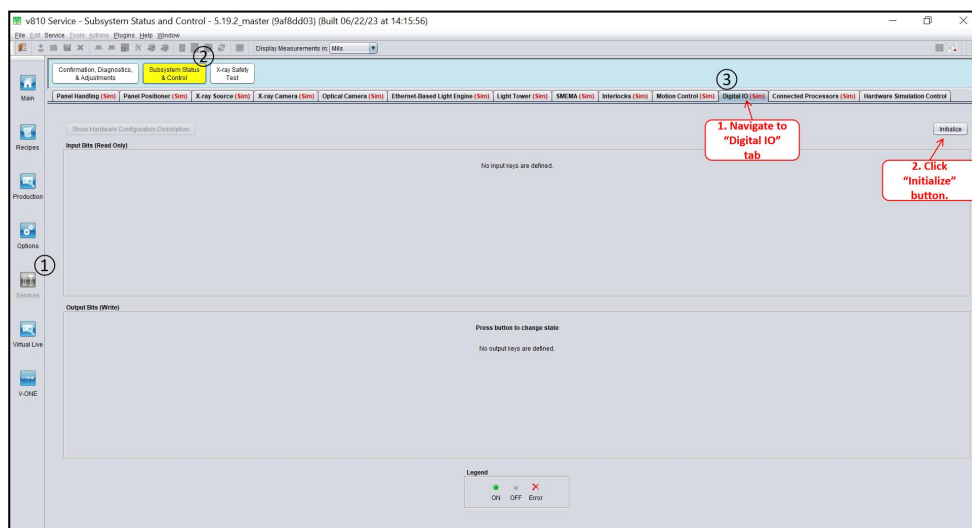


Figure 16: DIO Initialization (Sample Image)

20. **Panel Positioner Initialization:** Upon completion of DIO initialize, navigate to ‘Panel Positioner’ tab and click “Initialize”.

#IMPORTANT This step must be successfully executed before proceed to sensors adjustment.

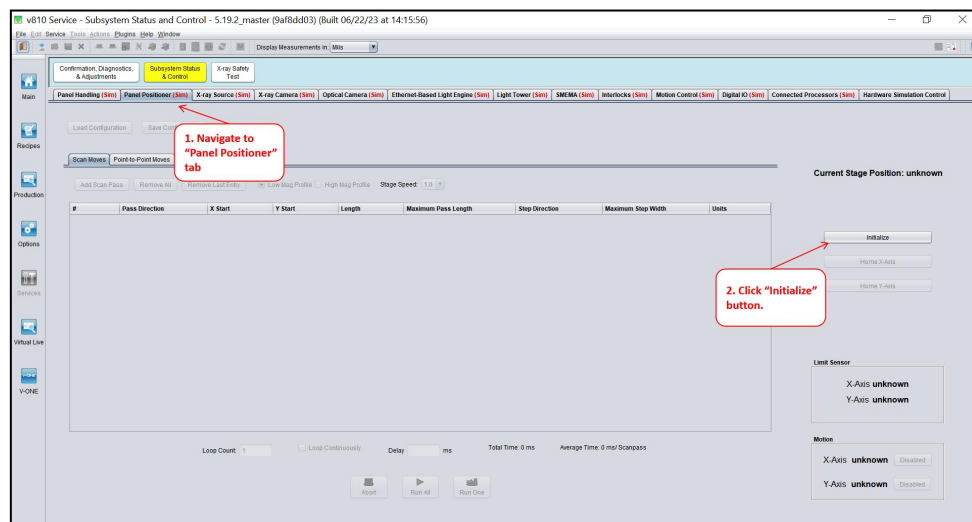


Figure 17: Initialize “Panel Positioner” (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors	Effective Date	30 Oct 2024
	Alignment Procedure		

21. **Verify the “Current Stage Position” for Y-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the Y stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, proceed to Step 22. Else, proceed to Step 25.

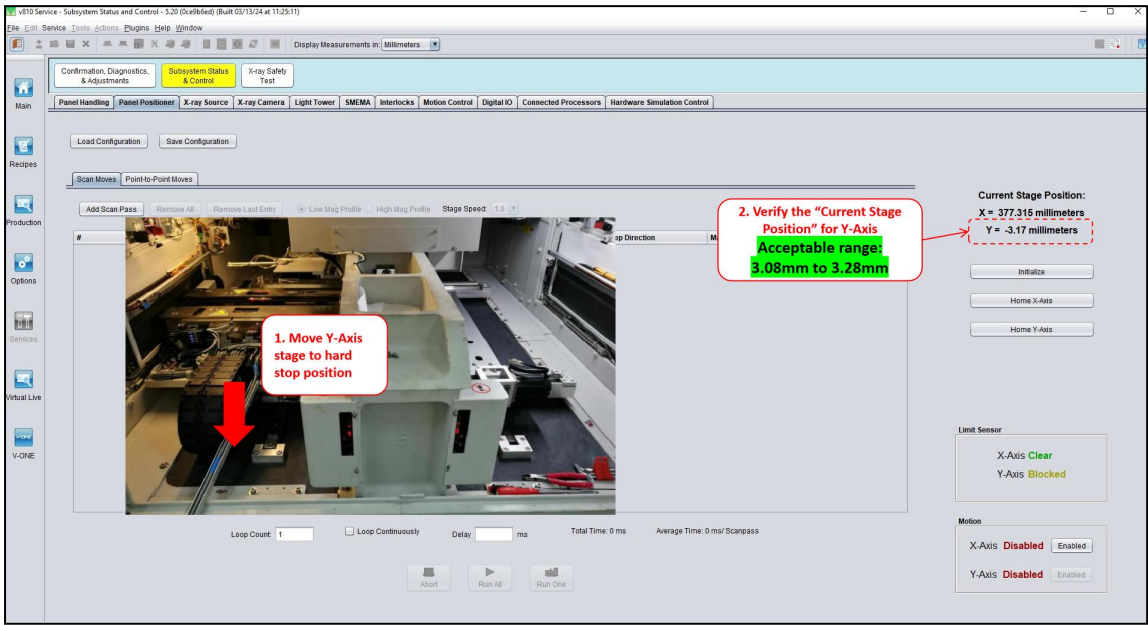


Figure 18: Current Stage Position for Y-Axis within range (Sample Image)

22. **Loosen the Screws for Spring Clamp and Limit Sensor:** Loosen all the screws for the spring clamp and limit sensor using precision tool. Be cautious not to damage the Quantic scale.

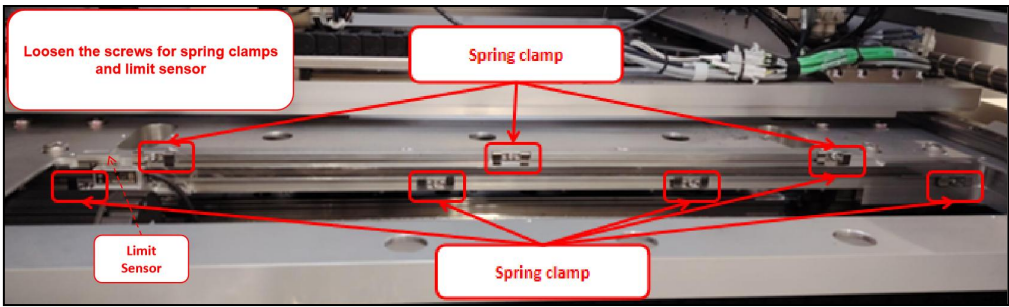


Figure 19: Quantic Scale Spring Clamps

23. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale as below table. While moving the Quantic scale, check the “Current Stage Position” in V810 and ensure the reading fall in the range of 3.08mm to 3.28mm.

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.08mm	Towards the front access door
More than -3.28mm	Towards the rear access door

Table 3: Y-Axis Quantic Scale Positioning Adjustment Guidelines

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

24. **Tighten the Screws for Spring Clamps and Limit Sensor:** Upon completion of Y-Axis home position alignment, tighten all screw for the spring clamps and limit sensor.

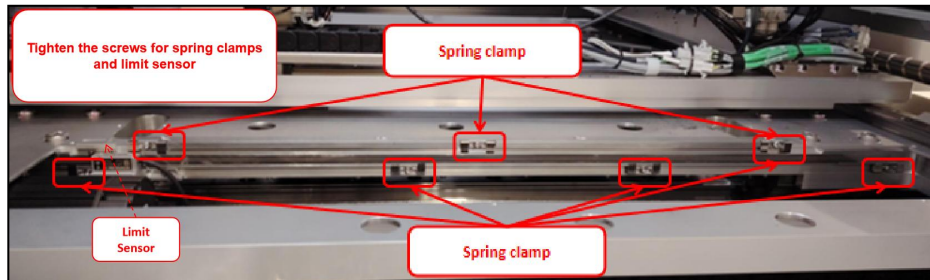


Figure 20: Y-Axis Quantic Scale Spring clamp

25. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the line of Quantic scale and stop at the right edge of the screw for the magnet sensor and tighten the screw.

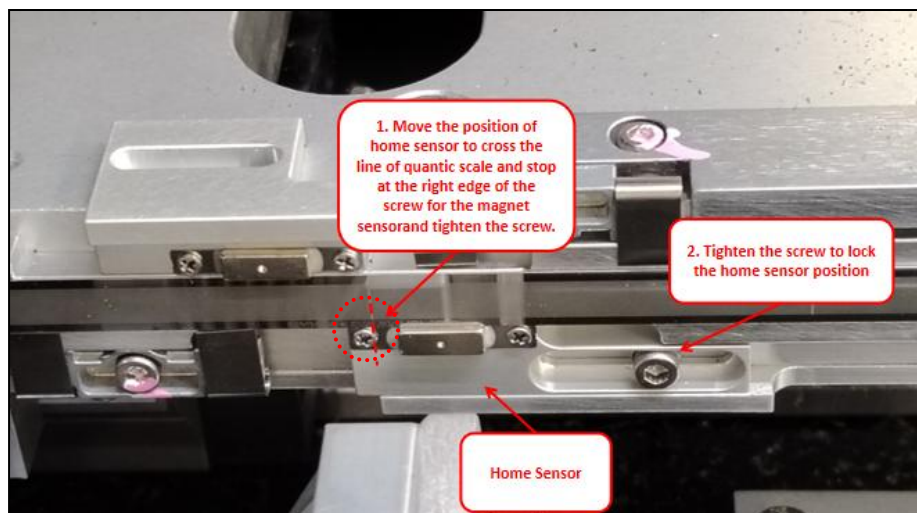


Figure 21: Y-Axis Home Sensor Offset

26. **Y-Axis Positioning (Away from Hard Stop):** Gently move the Y-Axis to the middle of the stage.

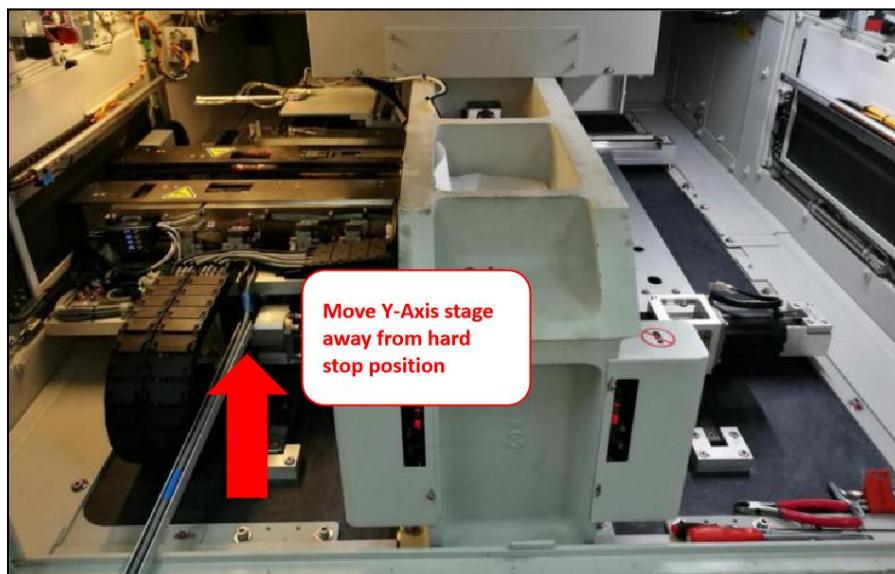


Figure 22: Move Y-Axis Stage Away From Hard Stop

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors	Effective Date	30 Oct 2024
	Alignment Procedure		

27. **Homing again:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for Y-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home Y-Axis”

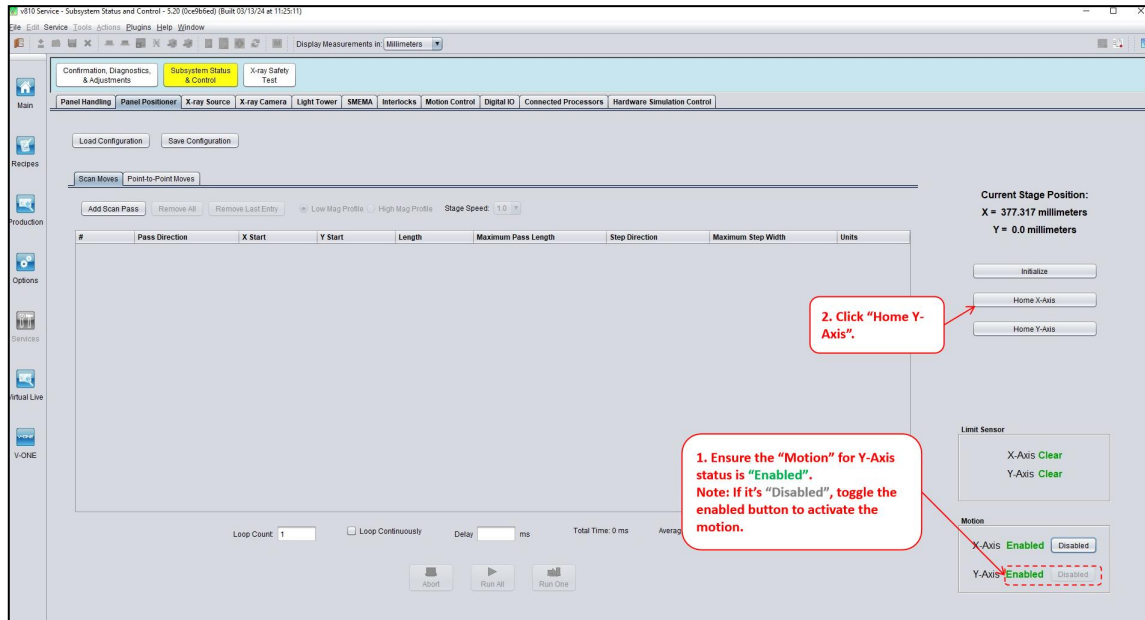


Figure 23: Limit Sensor Status Verification after Panel Positioner Initialize for Y-Axis

28. **Y-Axis “Current Stage Position” Verification (Hard Stop Position):** Move the Y-Axis stage to hard stop position, double confirm the current stage position is within the range of 3.08mm to 3.28mm. If current position of Y-Axis is out of range from 3.08mm to 3.28mm, route back to Step 22. Else, proceed to Step 29.

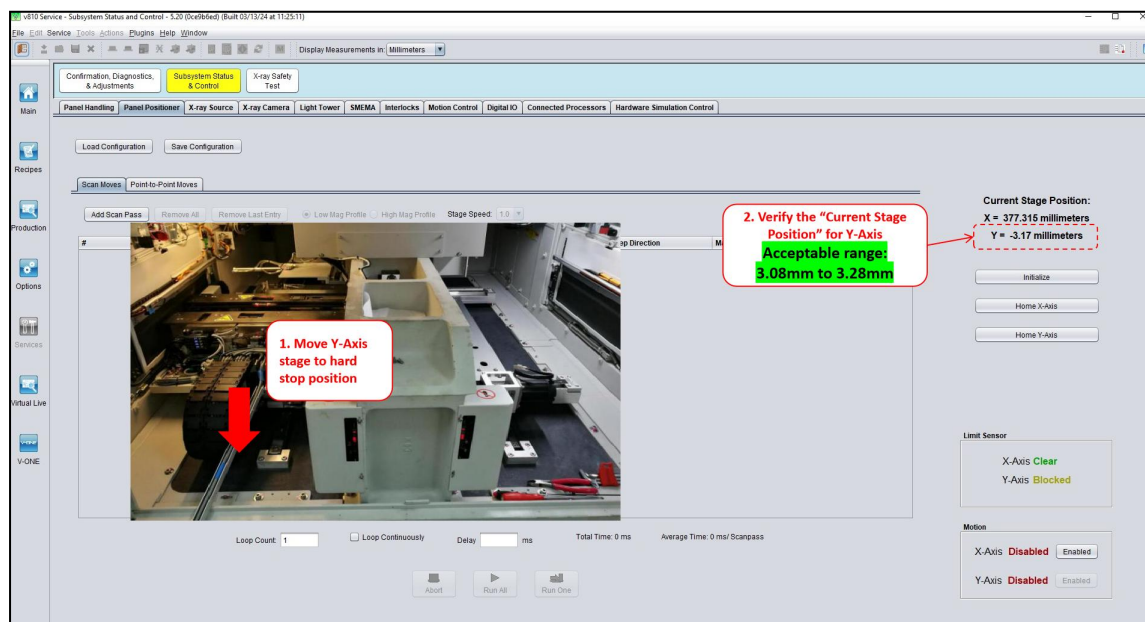


Figure 24: Verification of Current Stage Position for Y-Axis

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

29. **Move the Stage towards Hard Stop direction until it hit -0.5mm:** Move the stage towards the hard stop direction and ensure the current stage position is stop at -0.5mm.

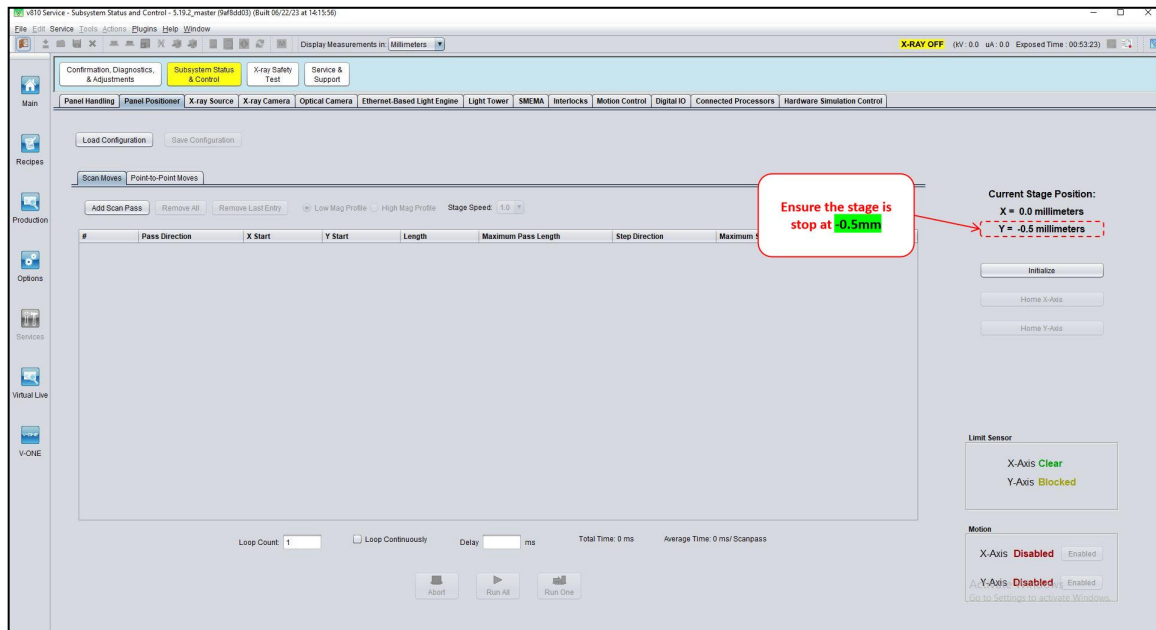


Figure 25: Move Y-Axis Stage to Current Stage Position -0.5mm (Sample Image)

30. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

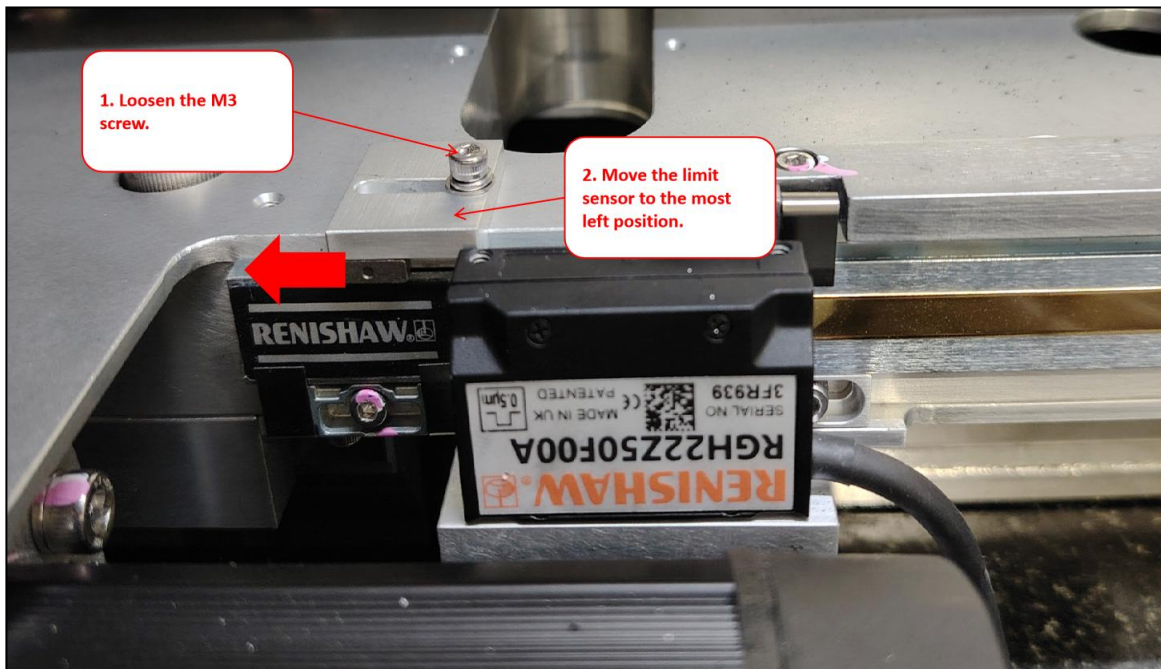


Figure 26: Move Limit Sensor to the Most Left Position

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

31. **Adjust the limit sensor to right hand side until the signal is “Blocked”:** Slowly move the limit sensor towards the right hand side position until the signal for limit sensor is blocked.

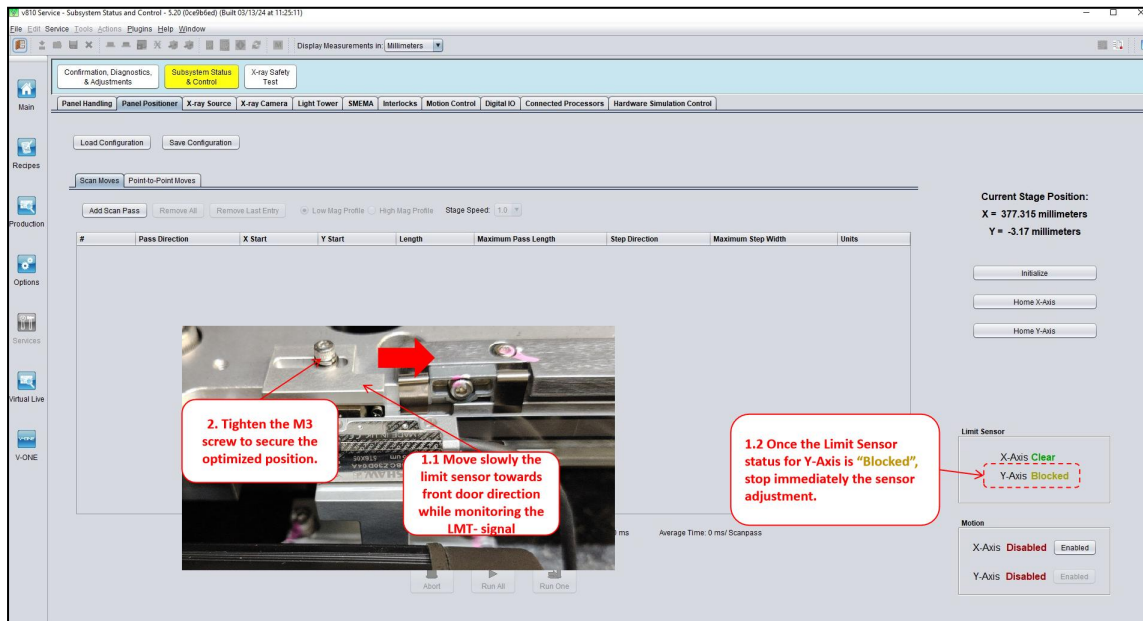


Figure 27: Positioning Limit Sensor to Triggered State

32. **Final Round Verification:** Lastly, manually move the X-stage to the center position, perform homing and ensure the signal for limit sensor is “Clear”. Then, manually move the stage to hard stop position and ensure the signal for limit sensor is “Blocked”.
33. **Y-Axis Cover Plate Assembly:** Carefully assemble back the Y-Axis cover plate
- Note:** If applicable.

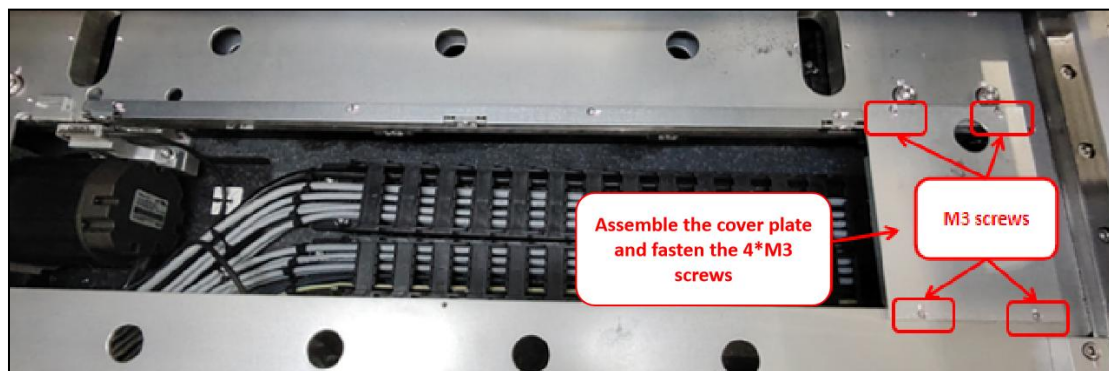


Figure 28: Y-Axis Cover Plate

34. **Validation of Machine Performance After Adjustment and Confirmation:** Machine performance validation is required before release back to customer.
- Note: If there is involvement of X-axis scale replacement, validation may be done upon all adjustments made.*

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

For Panasonic Motion Drive Only

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

X-Axis Home and Limit Sensor Alignment for Panasonic Drive

35. **Unplug the LAN Cable Labeled with Slice IO from the Atom PC:** Disconnect the LAN cable from Atom PC that is connecting to the slice IO.

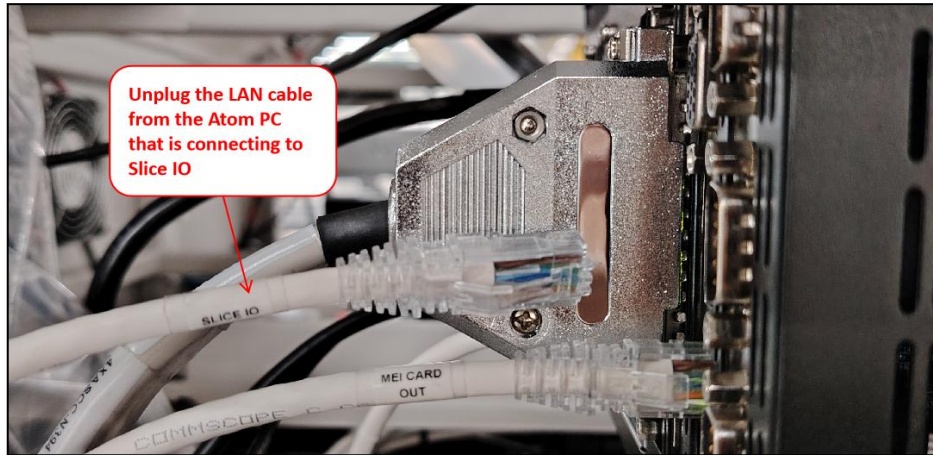


Figure 29: Unplug LAN cable from Atom PC

36. **Launch Remote Desktop Connection:** Go to search, key in remote desktop connection and launch the application.

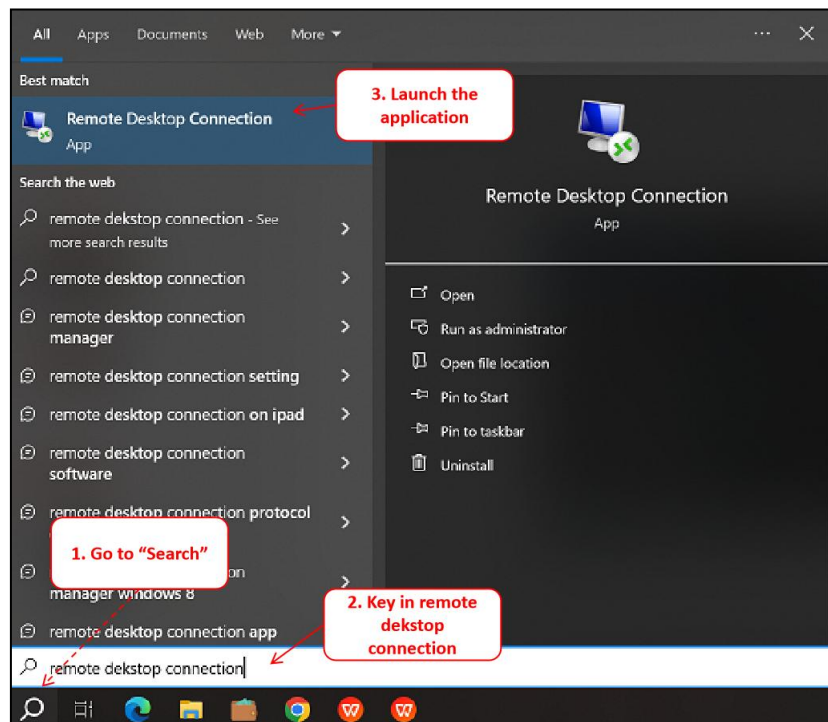


Figure 30: Launch the Remote Desktop Connection (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

37. **Connect to Atom PC (192.168.128.70):** Key in 192.168.128.70 at the computer address input box, hit enter to “Connect”.



Figure 31: Remote Desktop Connection (Sample Image)

38. **Launch the PanasonicHomeSensorAdjustmentTools.exe Program:** Navigate to Desktop and look for PanasonicHomeSensorAdjustmentTools.exe application. Double click the icon to open the program.

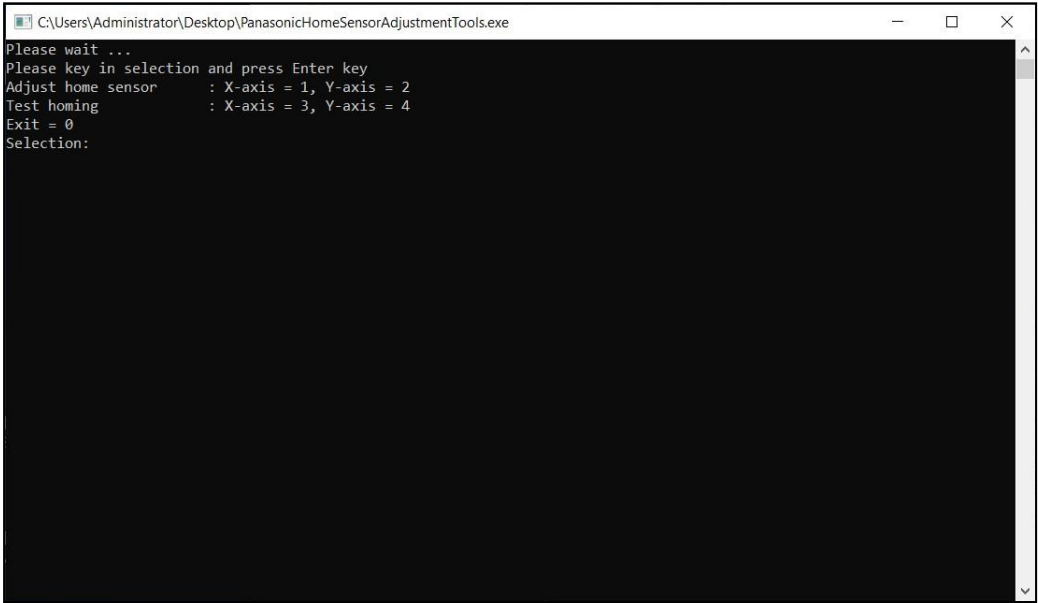


Figure 32: Launch PanasonicHomeSensorAdjustmentTools

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

39. **Launch the “Common Motion Utility” Application:** Go to Windows Search. Input “Common Motion Utility” and launch the application. Wait patiently for the application to startup.

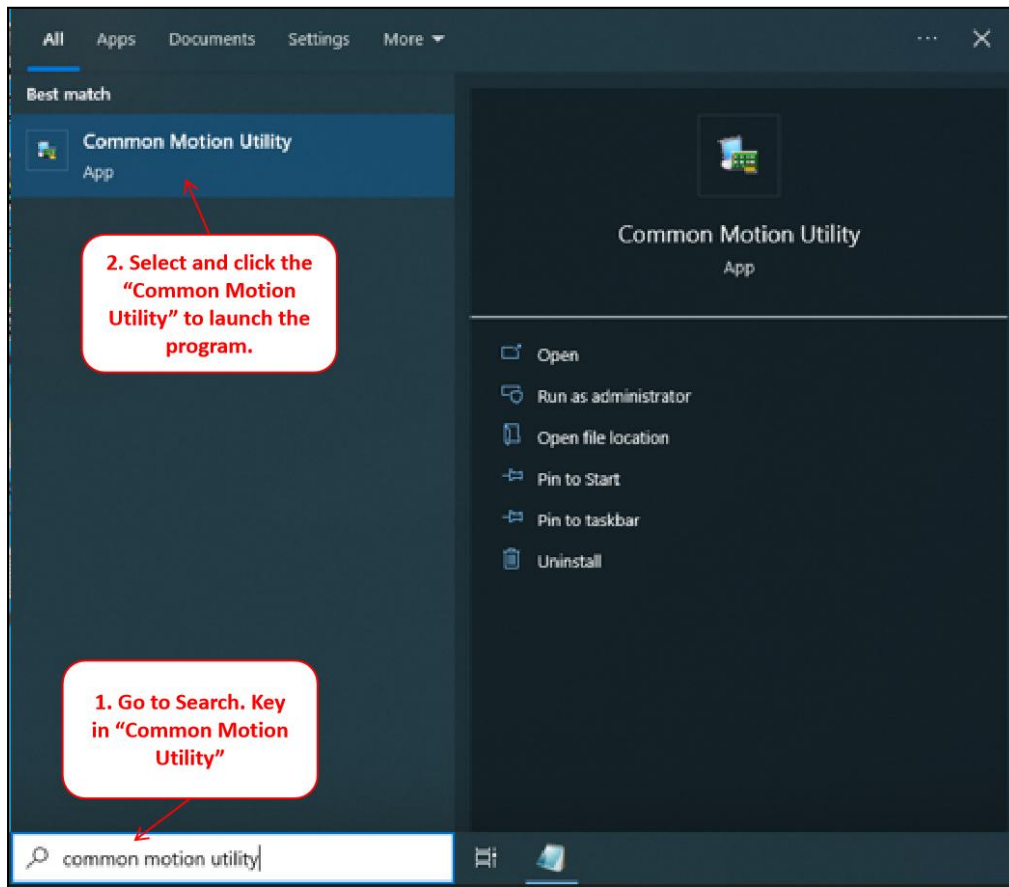


Figure 33: Launch Common Motion Utility Application

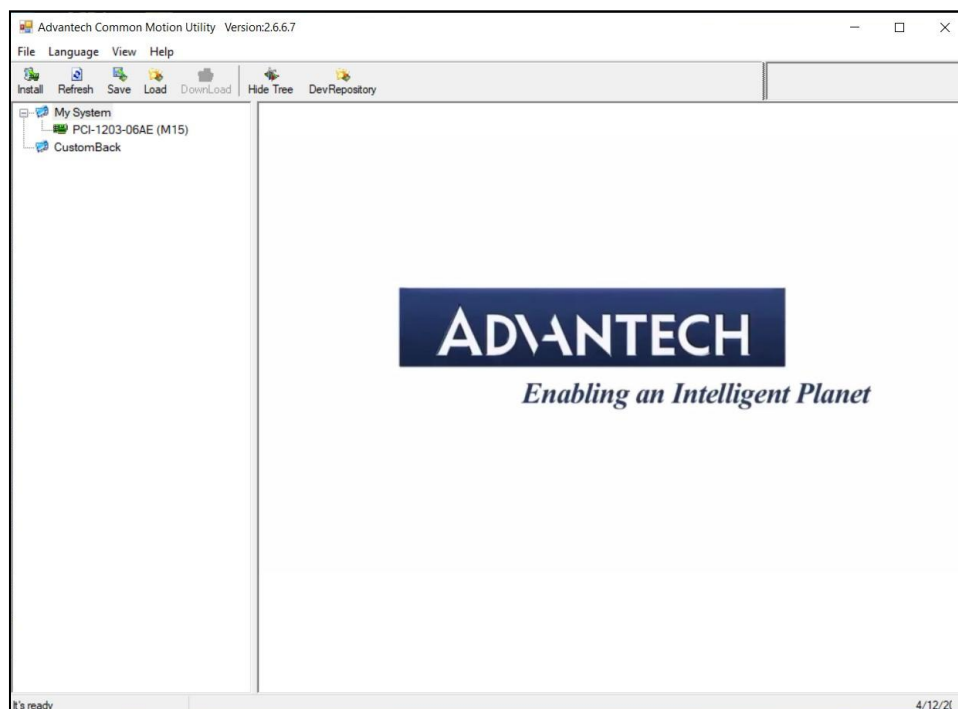


Figure 34: GUI for Common Motion Utility Startup

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

40. **Expand the list for PCI-1203-06AE (M15):** Double click the PCI-1203-06AE (M15) to expand its selection list. Under the Motion Ring node, there are 2 items as per below.

40.1 0x001: MCDLT35BF: For X-axis home position and limit sensor adjustment

40.2 0x002: MCDLT35BF: For Y-axis home position and limit sensor adjustment

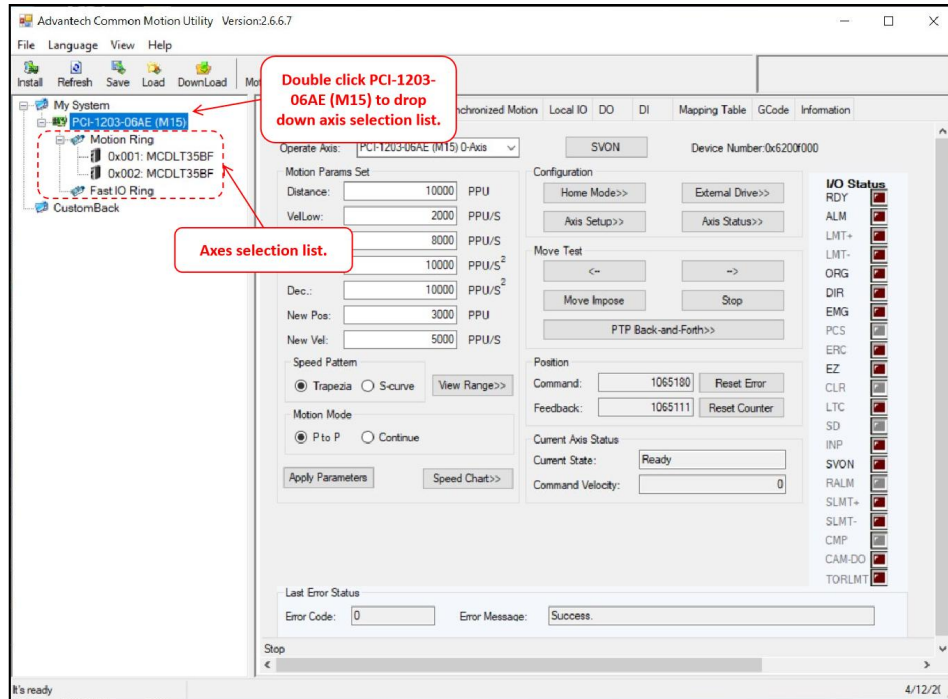


Figure 35: List Expansion for PCI-1203-06AE (M15)

41. **Axis Selection:** Select the 0x001: MCDLT35BF to proceed home position adjustment for x-axis.

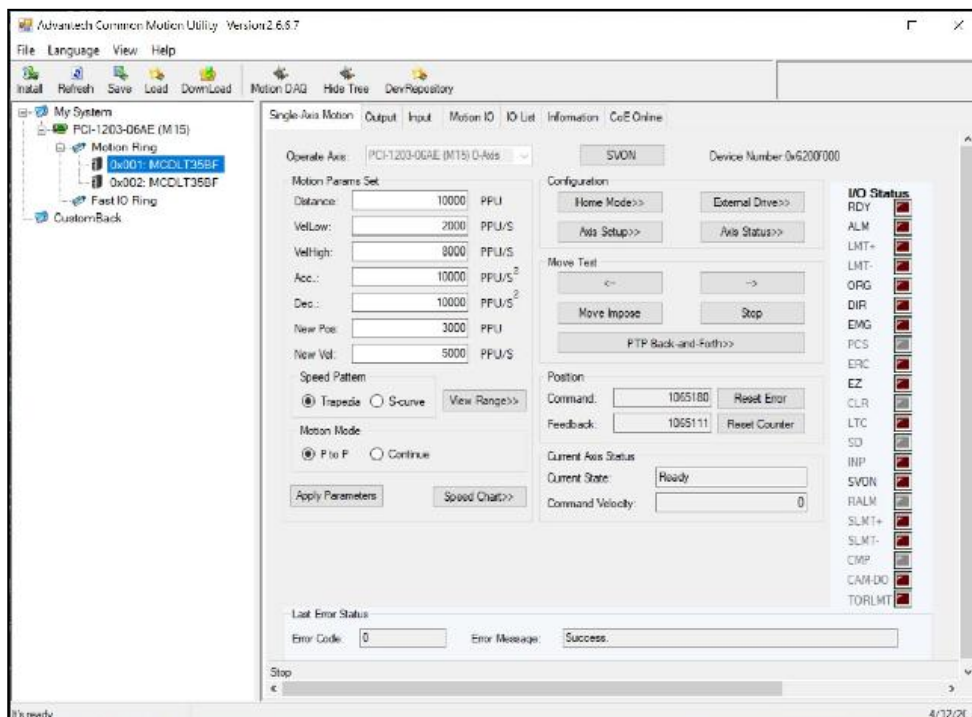


Figure 36: Axis Selection for Home Position Adjustment

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

42. **Ensure LMT- is Triggered at Hard Stop Position:** Upon axis selection, move the specific axis stage to hard stop position and ensure the LMT- is in triggered mode. Else, adjust the limit sensor for signal to be triggered.

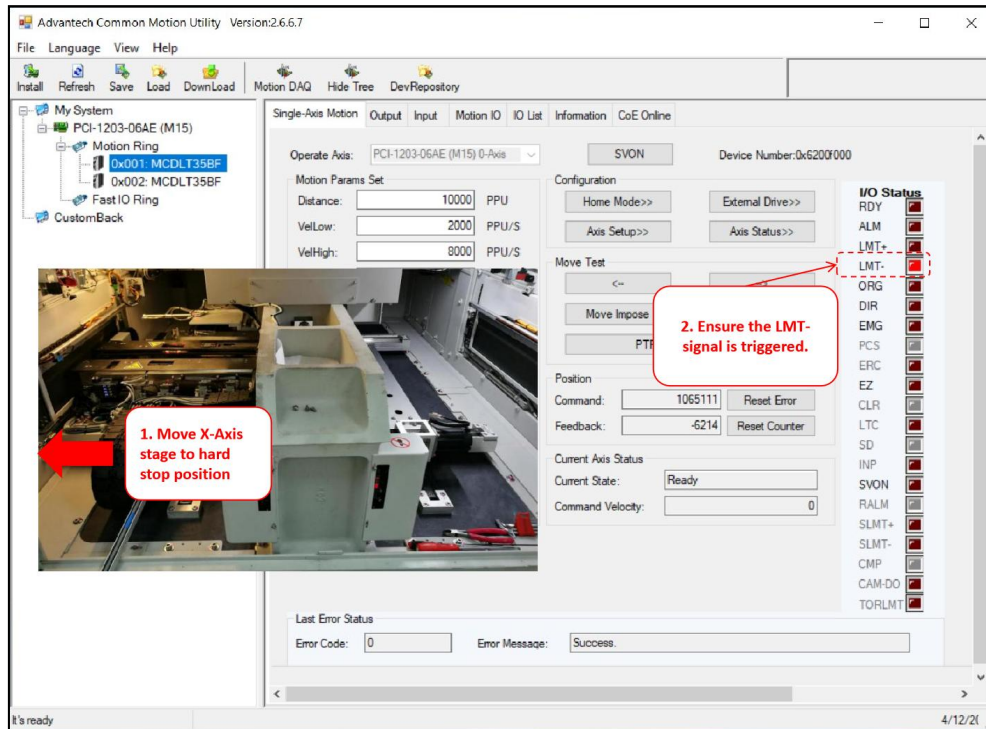


Figure 37: LMT- Signal Triggering Status

43. **X-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the X-Axis stage away from hard stop position.

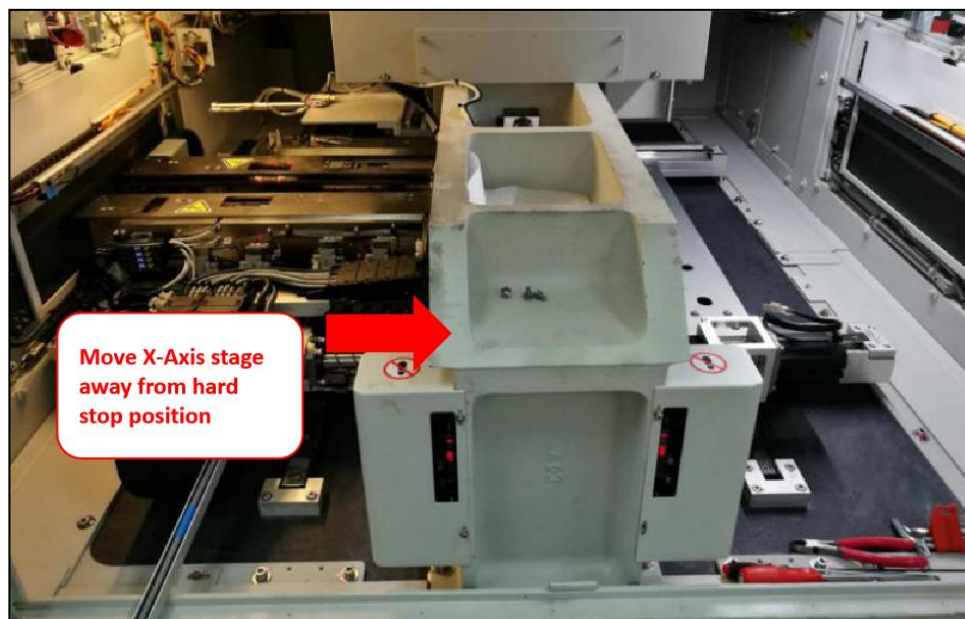


Figure 38: Move X-Axis Stage Away From Hard Stop

	Work Instruction AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Doc. Revision Effective Date	01 30 Oct 2024
--	---	--	--

44. **Perform Homing:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

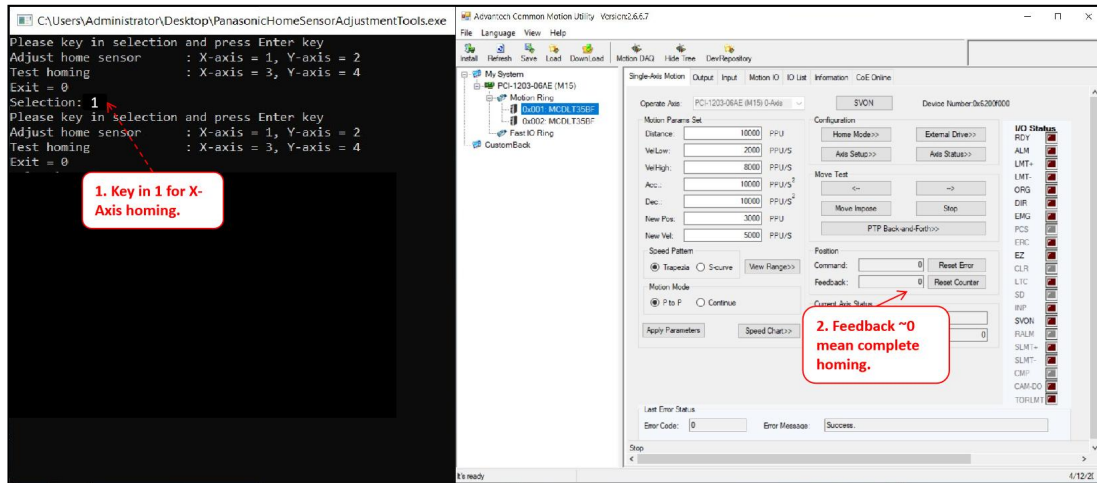


Figure 39: Perform Homing

45. **Move the Stage to Hard Stop Position:** Move the X-stage to the hard stop position.

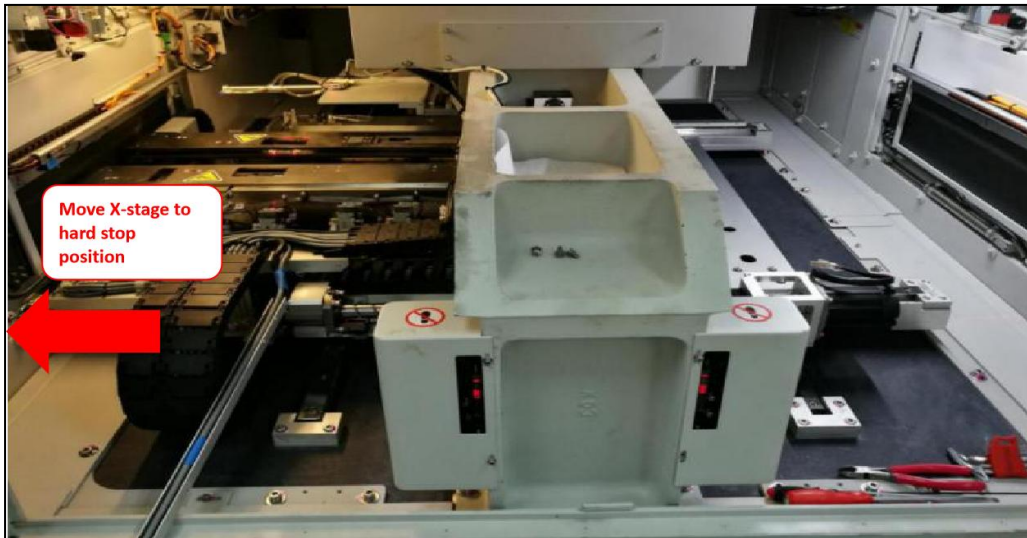


Figure 40: Move Stage to Hard Stop Position

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

46. **Validate the Feedback Position is 4980-5180:** At the hard stop position, validate the feedback position. If it is not within acceptable range of 4980-5180, please proceed with Step 48. Else, proceed to Step 51.

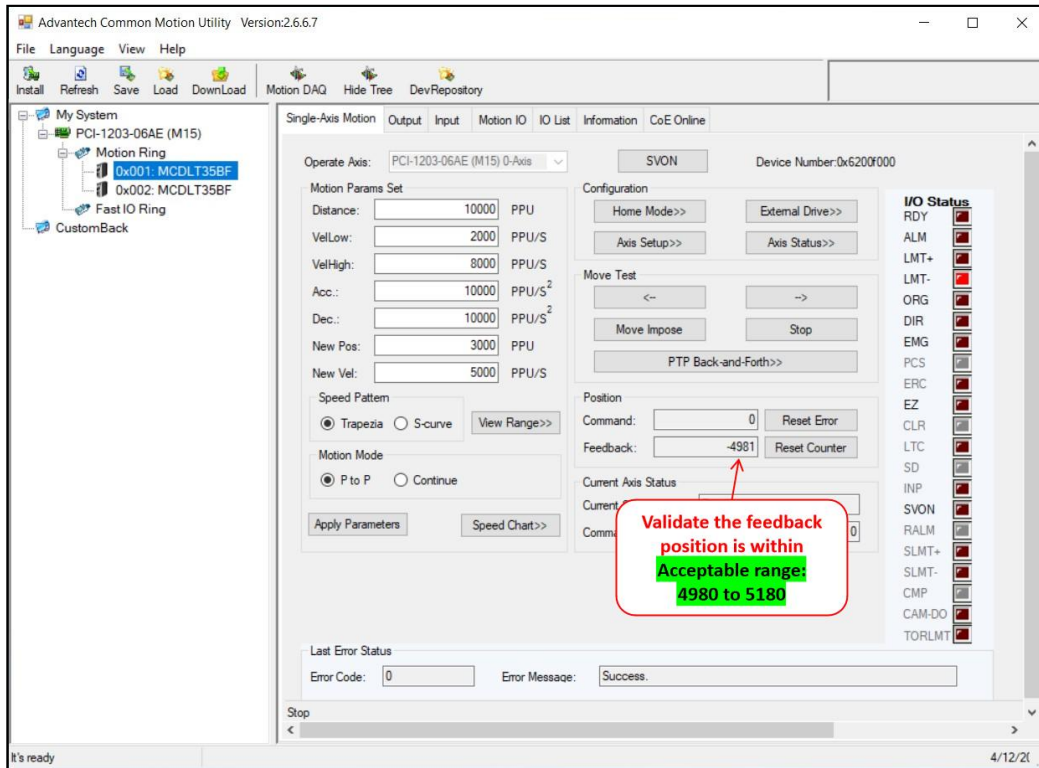


Figure 41: Validation of Feedback Position

47. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp.

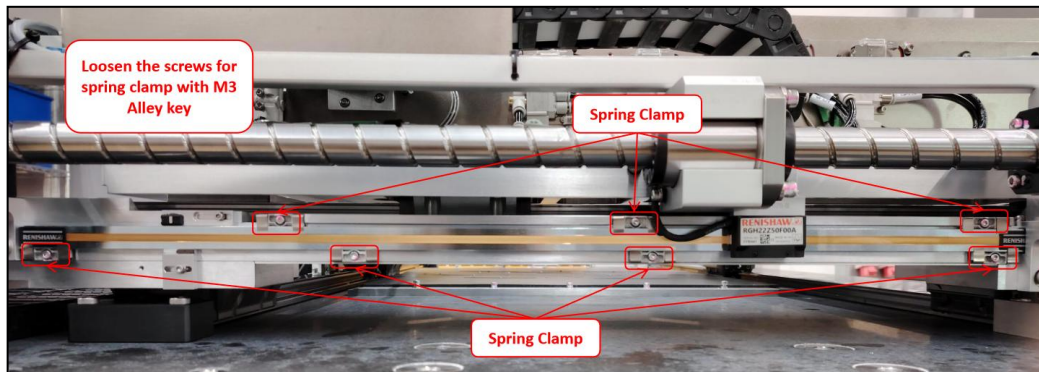


Figure 42: Quantic Scale Spring Clamps (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

48. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale. While moving the Quantic scale, check the “Feedback Position” in Common Motion Utility and ensure the reading fall in the range 4980 to 5180.

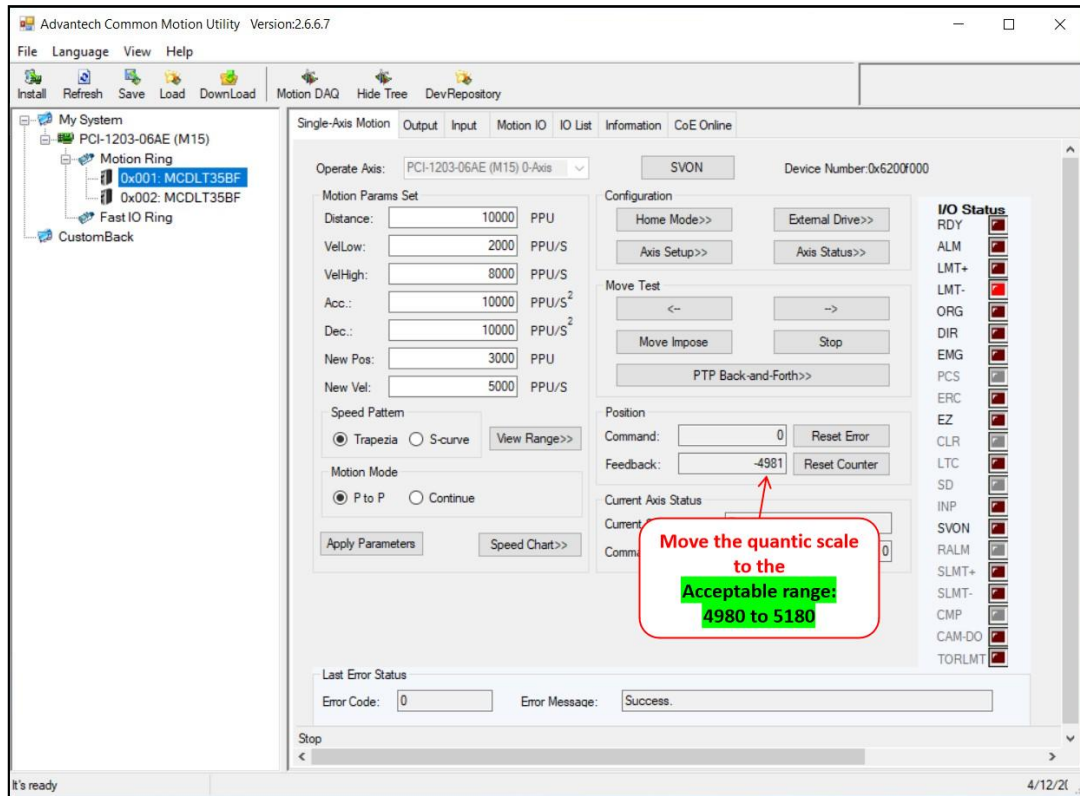


Figure 43: Quantic Scale Adjustment by Monitoring the Feedback Position

49. **Tighten the Spring Clamps:** Upon completion of home position alignment, tighten all screw for the spring clamps.

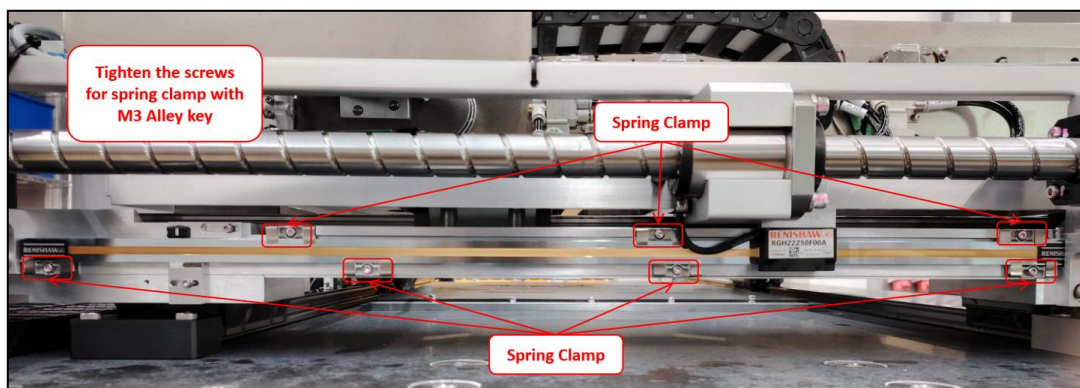


Figure 44: X-Axis Quantic Scale Spring clamp (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

50. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the engraved line mark at the top side of the Quantic scale.

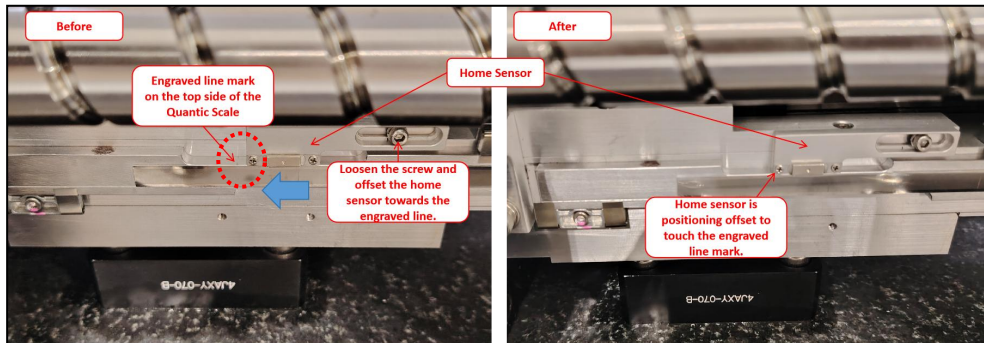


Figure 45: X-Axis Home Sensor Offset

51. **X-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the X-Axis to the middle of the stage.

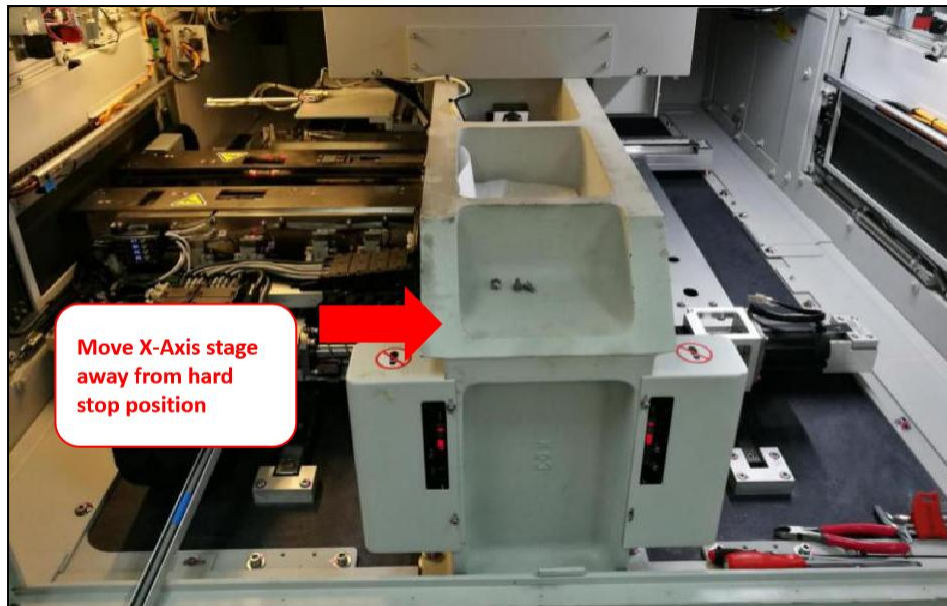


Figure 46: Move X-Axis Stage Away From Hard Stop

	Work Instruction AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Doc. Revision	01
		Effective Date	30 Oct 2024

52. **Homing again:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

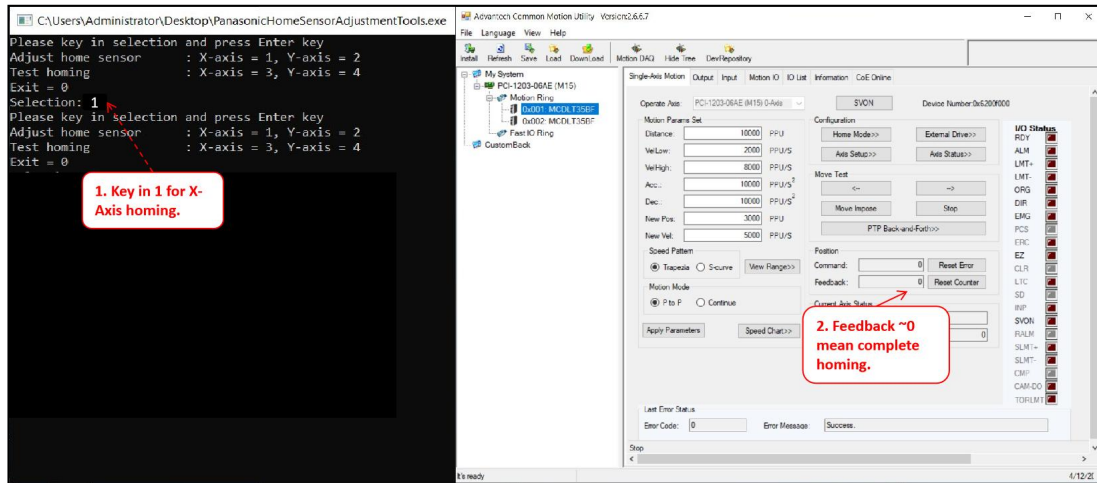


Figure 47: Perform Homing

53. **Move the Stage towards Hard Stop direction until it hit 1755 ± 1 :** Move the stage towards the hard stop direction and ensure the feedback position is within the range of 1754 to 1756.

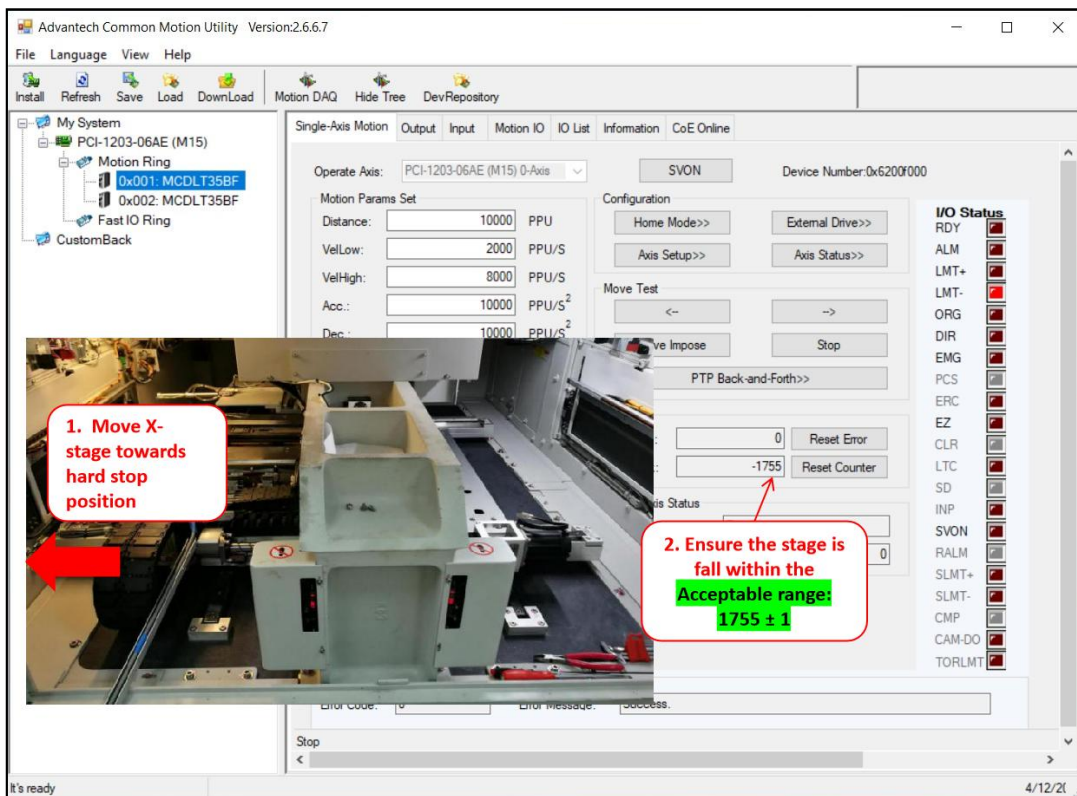


Figure 48: Move X-Axis Stage to Feedback Position 1755 (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

54. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

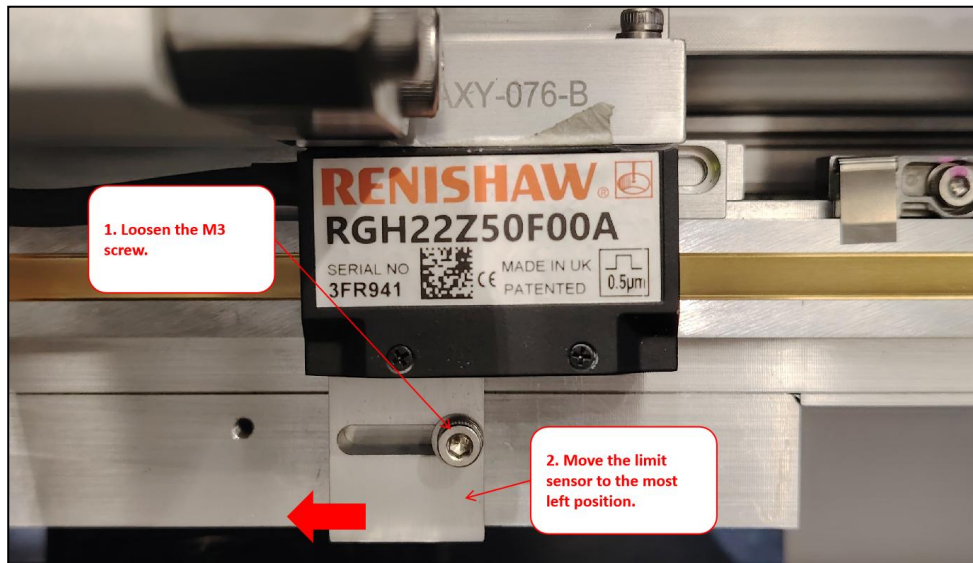


Figure 49: Move Limit Sensor to the Most Left Position

55. **Adjust the limit sensor to right hand side until LMT- is triggered:** Slowly move the limit sensor towards the right hand side position until the LMT- signal is triggered.

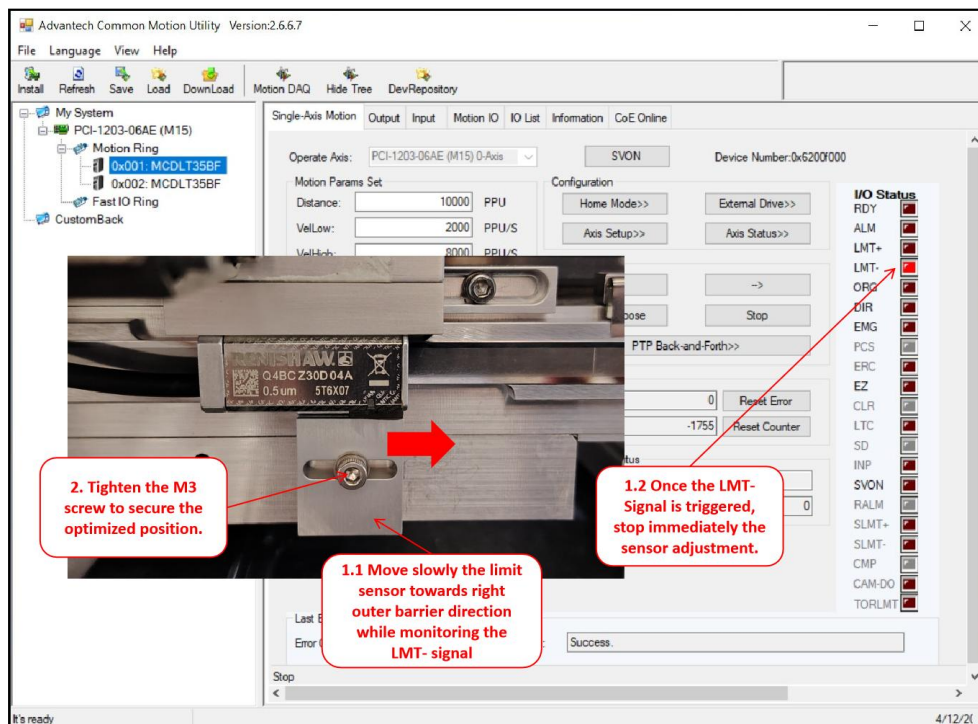


Figure 50: Positioning Limit Sensor to Triggered State

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

56. **Proceed to Test Homing for Final Verification:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 3 for x-axis to test homing result.

Note: If error persist, close all the application and route back to Step 39.

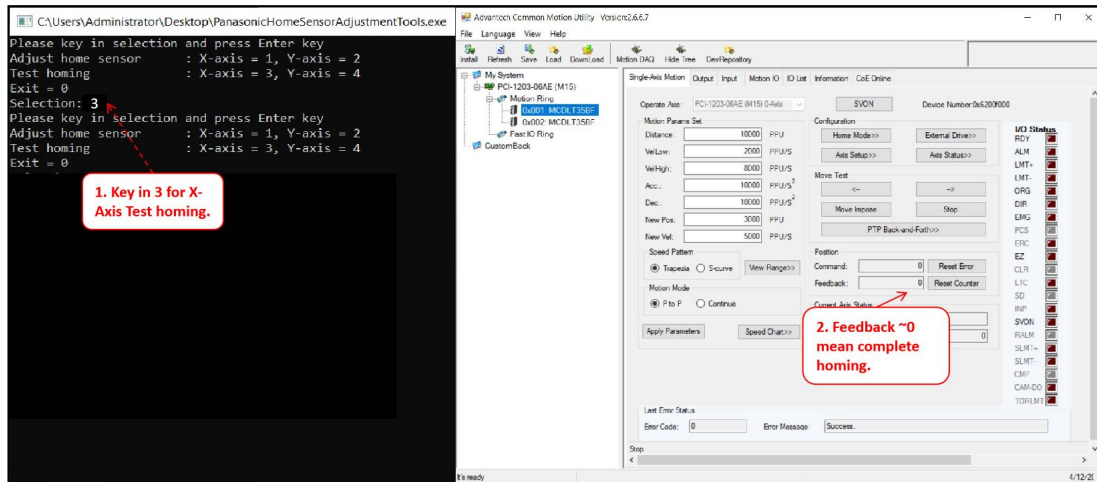


Figure 51: Perform Homing for Final Verification

57. **Re-connect the LAN cable labeled with Slice IO to the Atom PC:** Re-connect the LAN cable labeled with Slice IO to the Atom PC.



Figure 52: Reconnect the LAN Cable for Slice IO

58. **Go to Main Server and open v810:** Close remote desktop connection and launch the v810 program.
59. **Panel Positioner Initialization:** Navigate to 'Service > Subsystem Status & Control > Panel Positioner'. Click "Initialize".

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

60. **Verify the “Current Stage Position” for X-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the X stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, route back to Step 35. Else, proceed to the next step.

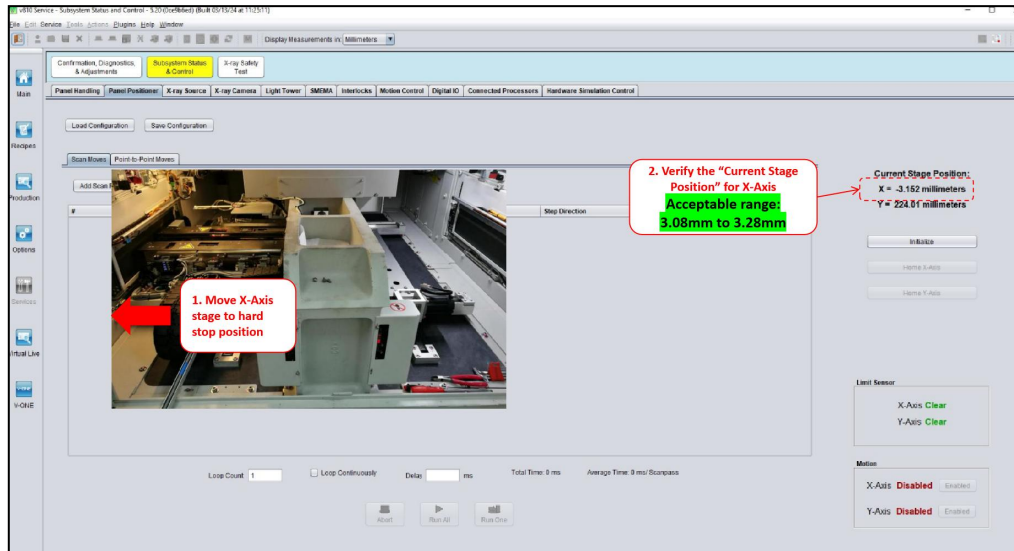


Figure 53: Verification of Current Stage Position for X-Axis

61. **Proceed to Validate Machine Performance:** Validation of machine if required before release the machine to production.

Note: If there is involvement of Y-axis scale replacement, validation may be done upon all adjustments made.

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

Y-Axis Home and Limit Sensor Alignment for Panasonic Drive

62. **Unplug the Slice IO LAN Cable from Atom PC:** Disconnect the LAN cable that connected to slice IO from Atom PC.

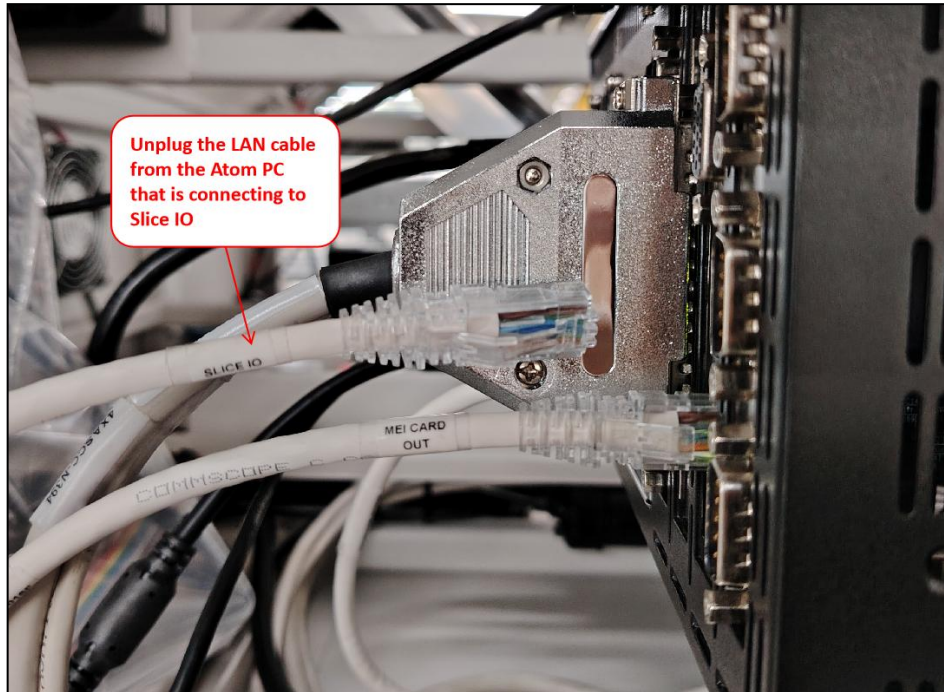


Figure 54: Unplug LAN Cable from the Atom PC

63. **Launch Remote Desktop Connection:** Go to search, key in remote desktop connection and launch the application.

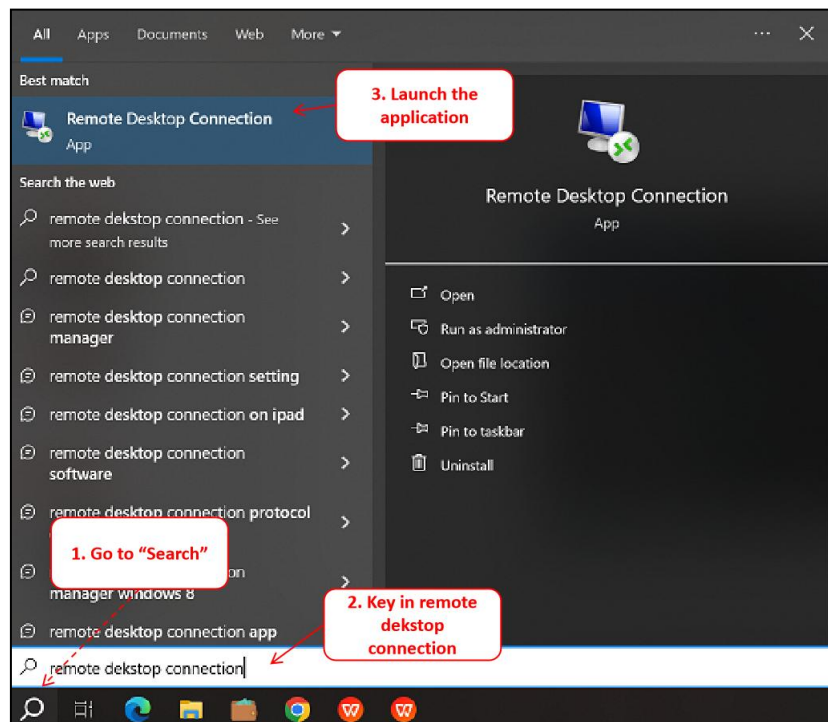


Figure 55: Launch the Remote Desktop Connection (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

64. **Connect to Atom PC (192.168.128.70):** Key in 192.168.128.70 at the computer address input box, hit enter to “Connect”.



Figure 56: Remote Desktop Connection (Sample Image)

65. **Launch the PanasonicHomeSensorAdjustmentTools.exe Program:** Navigate to Desktop and look for PanasonicHomeSensorAdjustmentTools.exe application. Double click the icon to open the program.

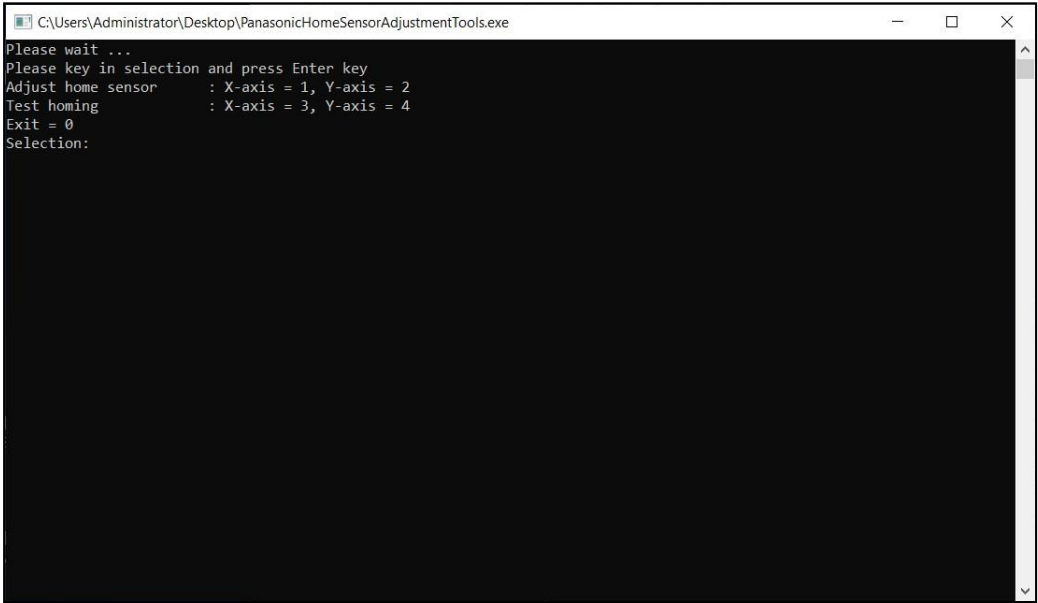


Figure 57: Launch PanasonicHomeSensorAdjustmentTools

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

66. **Launch the “Common Motion Utility” Application:** Go to Windows Search. Input “Common Motion Utility” and launch the application. Wait patiently for the application to startup.

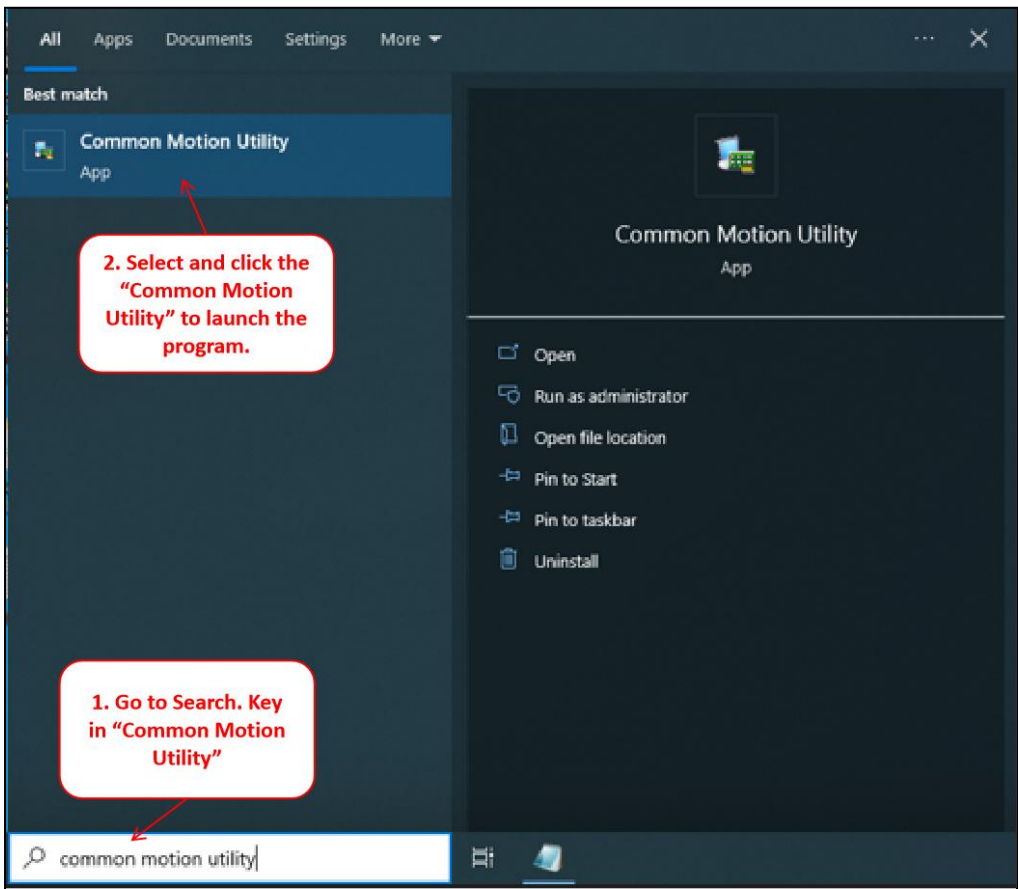


Figure 58: Launch Common Motion Utility Application

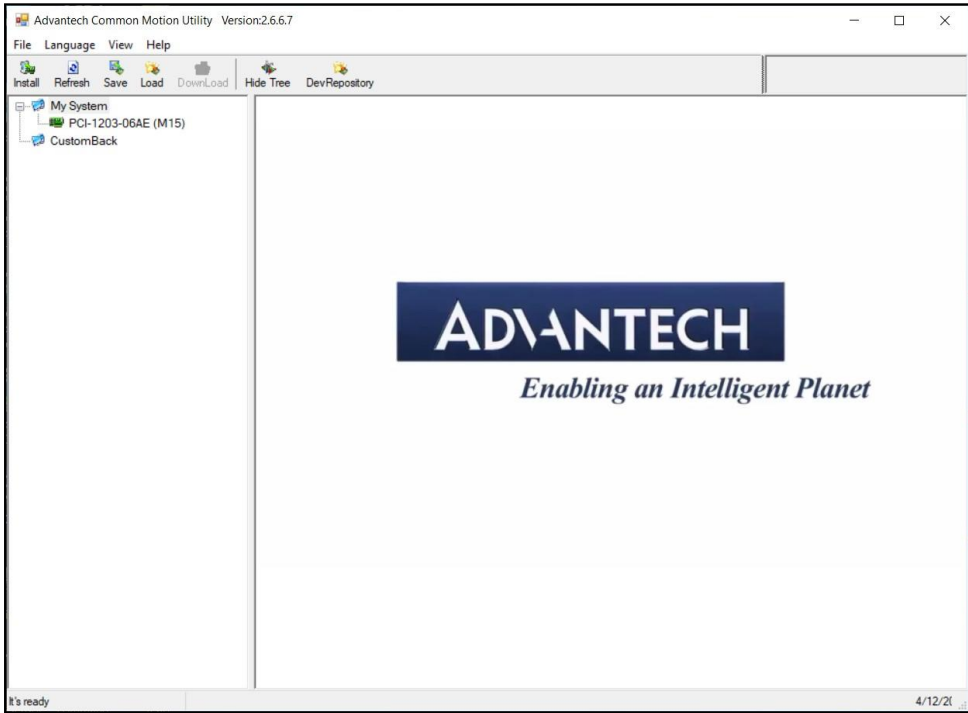


Figure 59: GUI for Common Motion Utility Startup

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

67. **Expand the list for PCI-1203-06AE (M15):** Double click the PCI-1203-06AE (M15) to expand its selection list. Under the Motion Ring node, there are 2 items as per below.

67.1 0x001: MCDLT35BF: For X-axis home position and limit sensor adjustment

67.2 0x002: MCDLT35BF: For Y-axis home position and limit sensor adjustment

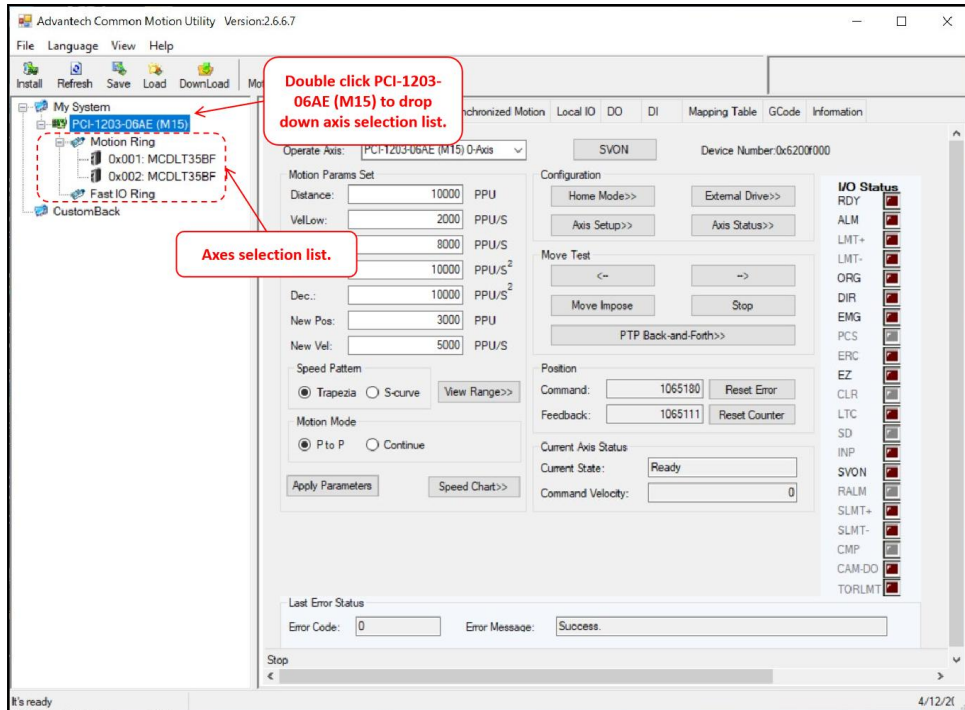


Figure 60: List Expansion for PCI-1203-06AE (M15)

68. **Axis Selection:** Select the 0x002: MCDLT35BF to proceed home position adjustment for x-axis.

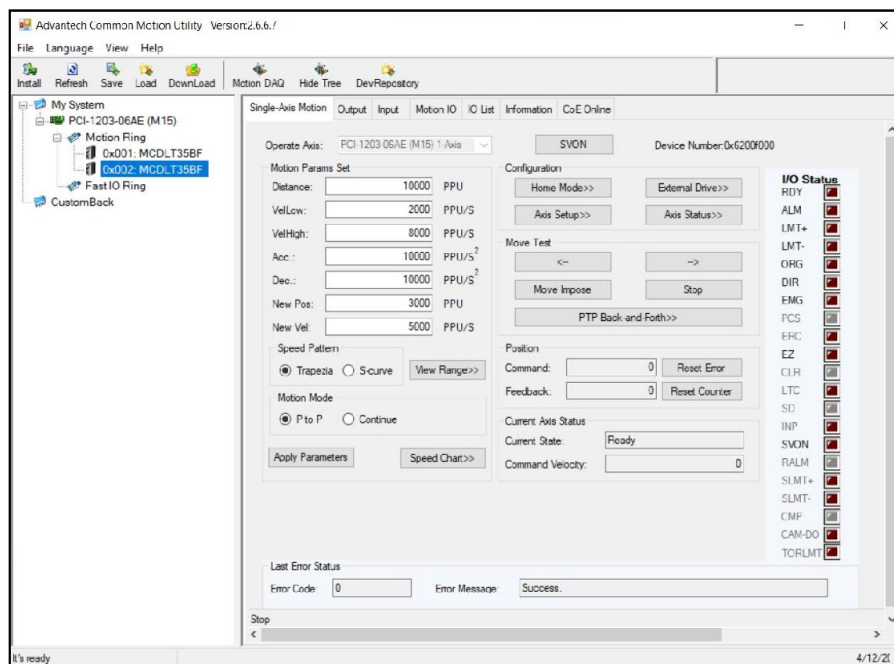


Figure 61: Axis Selection for Home Position Adjustment

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

69. **Ensure Limit - is Enabled at Hard Stop Position:** Upon axis selection, move the specific axis stage to hard stop position and ensure the limit - is in triggered mode. Else, adjust the limit sensor for signal to be triggered.

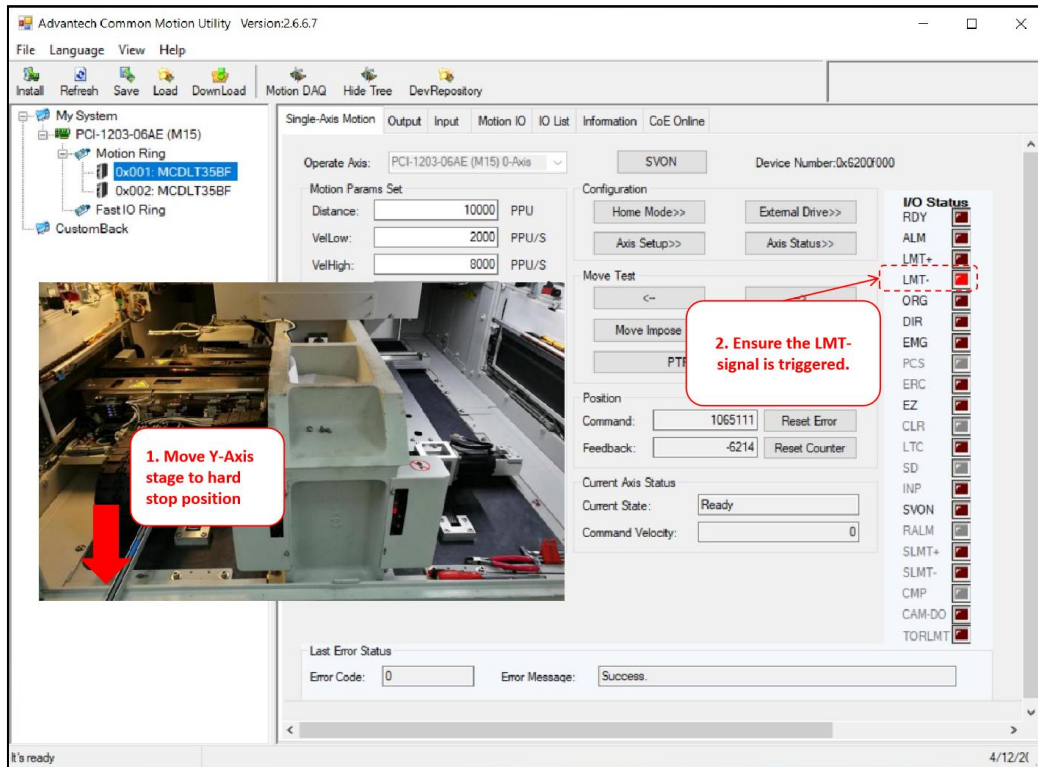


Figure 62: LMT- Signal Triggering Status

70. **Y-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the Y-Axis stage away from hard stop position.

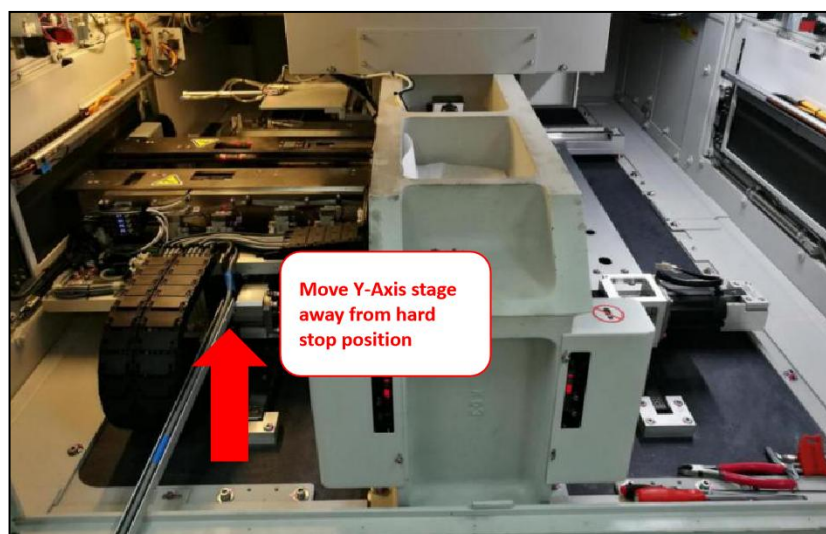


Figure 63: Move Y-Axis Stage Away From Hard Stop

	Work Instruction AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Doc. Revision	01
		Effective Date	30 Oct 2024

71. **Perform Homing:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 2 for y-axis to go home position.

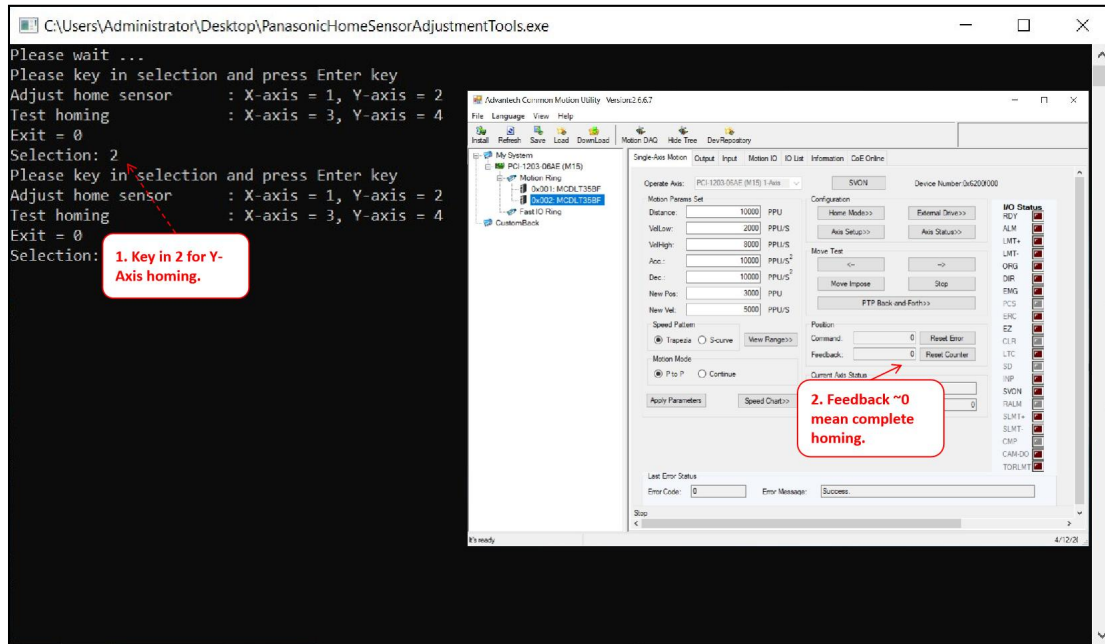


Figure 64: Perform Homing

72. **Move the Y-Stage to Hard Stop Position:** Move the Y-stage to the hard stop position.

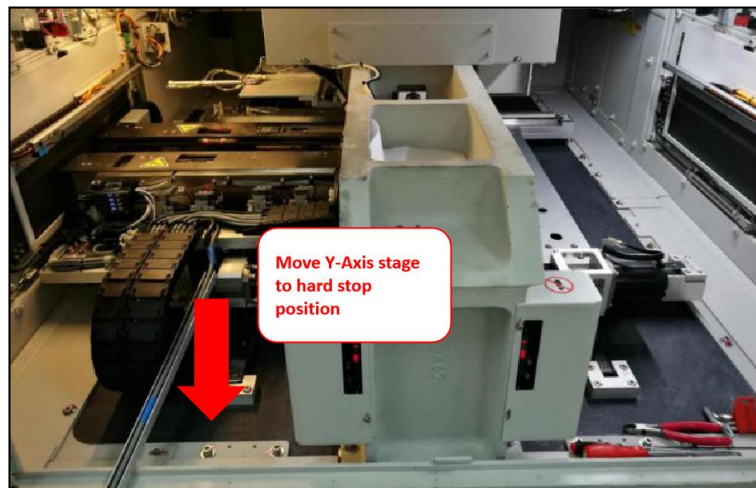


Figure 65: Move Y-Stage to Hard Stop Position

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors	Effective Date	30 Oct 2024
	Alignment Procedure		

73. **Validate the Feedback Position is 4980-5180:** At the hard stop position, validate the feedback position. If it is not within acceptable range of 4980-5180, please proceed with Step 77. Else, proceed to Step 80.

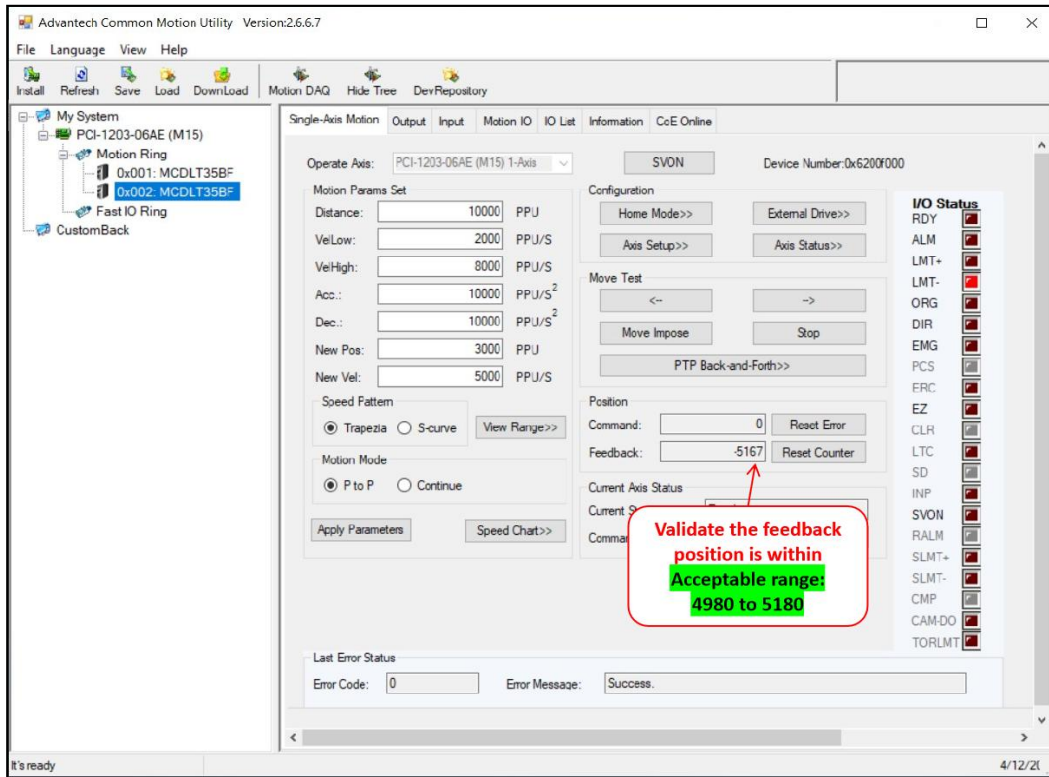


Figure 66: Feedback Position Validation

74. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp.

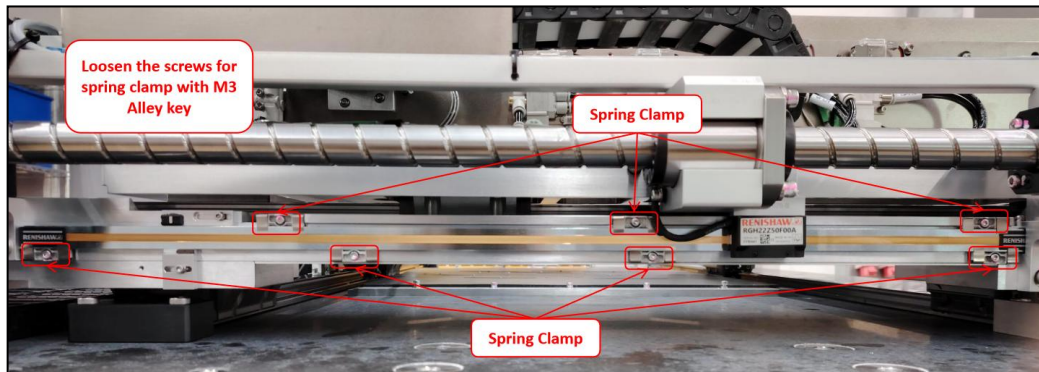


Figure 67: Quantic Scale Spring Clamps (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

75. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale. While moving the Quantic scale, check the “Feedback Position” in Common Motion Utility and ensure the reading fall in the range 4980 to 5180.

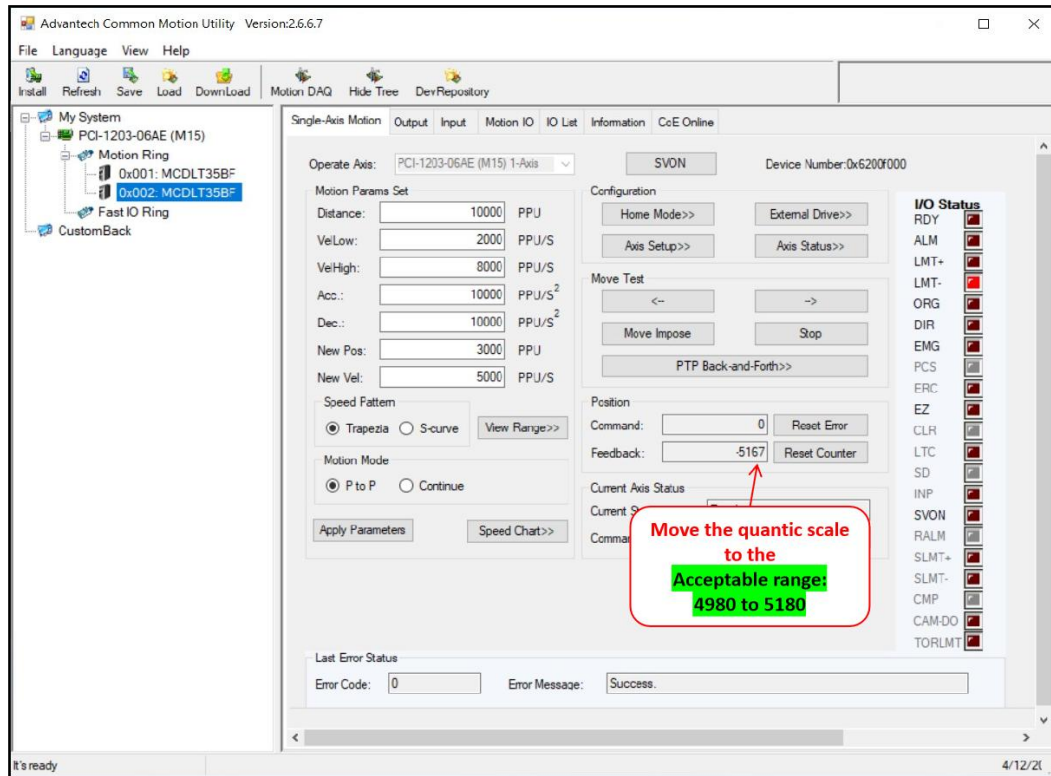


Figure 68: Quantic Scale Adjustment by Monitoring the Feedback Position

76. **Tighten the Spring Clamps:** Upon completion of home position alignment, tighten all screw for the spring clamps.

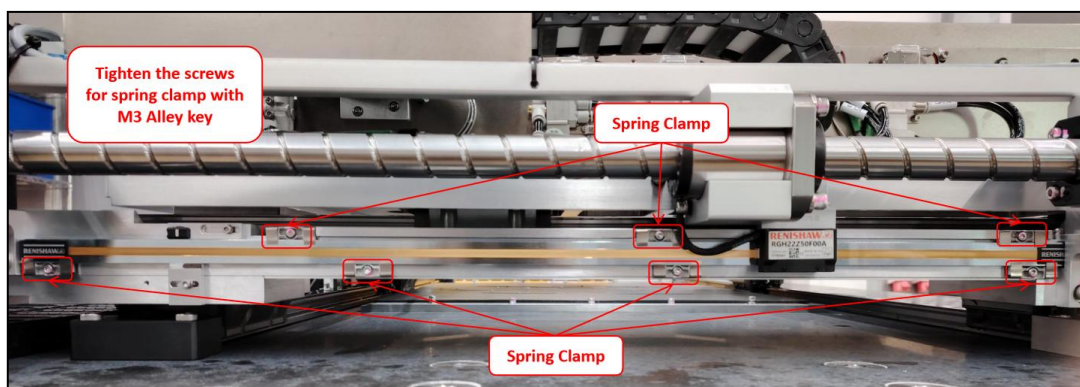


Figure 69: Y-Axis Quantic Scale Spring clamp (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

77. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the engraved line mark at the top side of the Quantic scale.

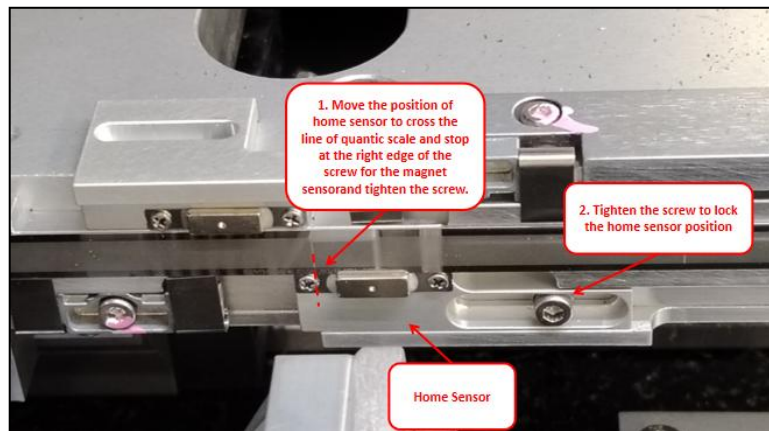


Figure 70: Y-Axis Home Sensor Offset

78. **Y-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the Y-Axis to the middle of the stage.

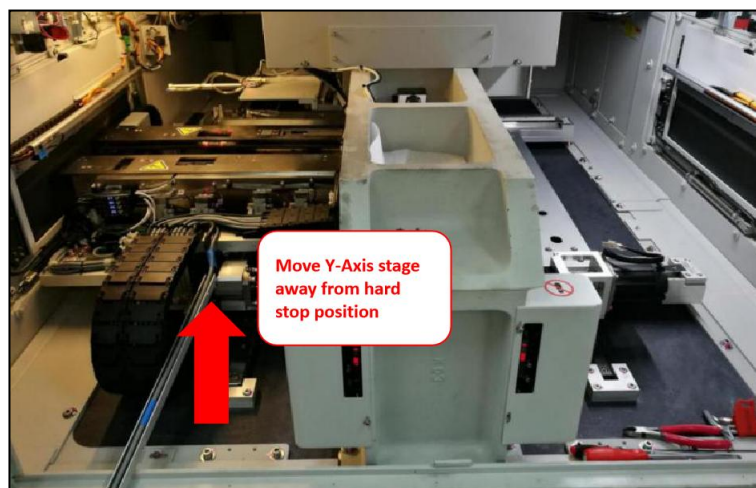


Figure 71: Move Y-Axis Stage Away From Hard Stop

	Work Instruction AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Doc. Revision	01
		Effective Date	30 Oct 2024

79. **Homing again:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

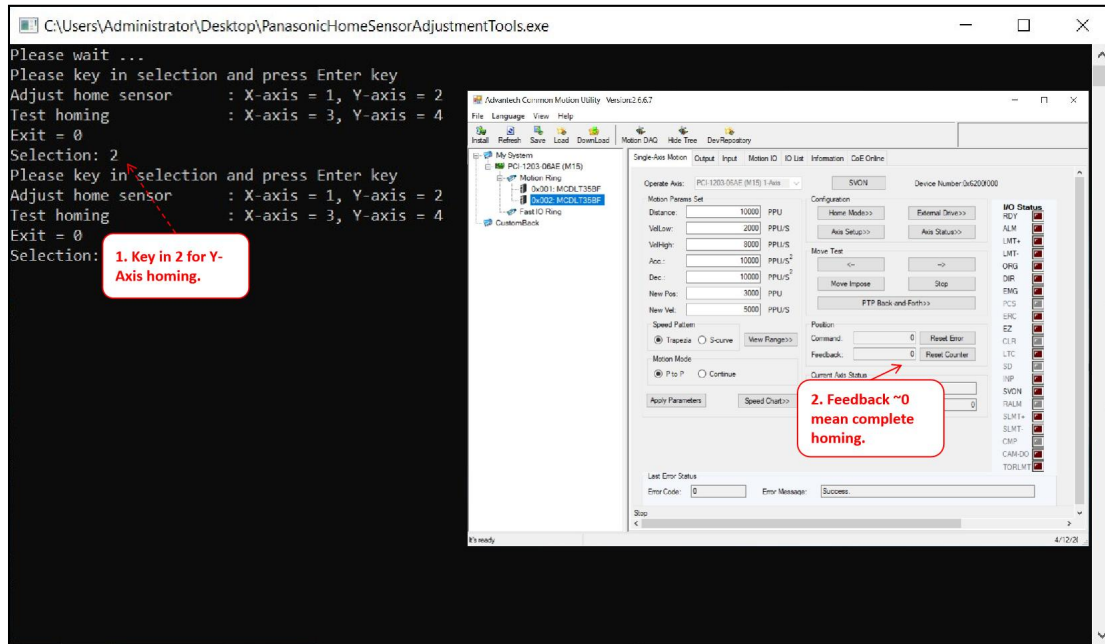


Figure 72: Perform Homing

80. **Move the Stage towards Hard Stop direction until it hit 1755 ± 1 :** Move the stage towards the hard stop direction and ensure the feedback position is within the range of 1754 to 1756.

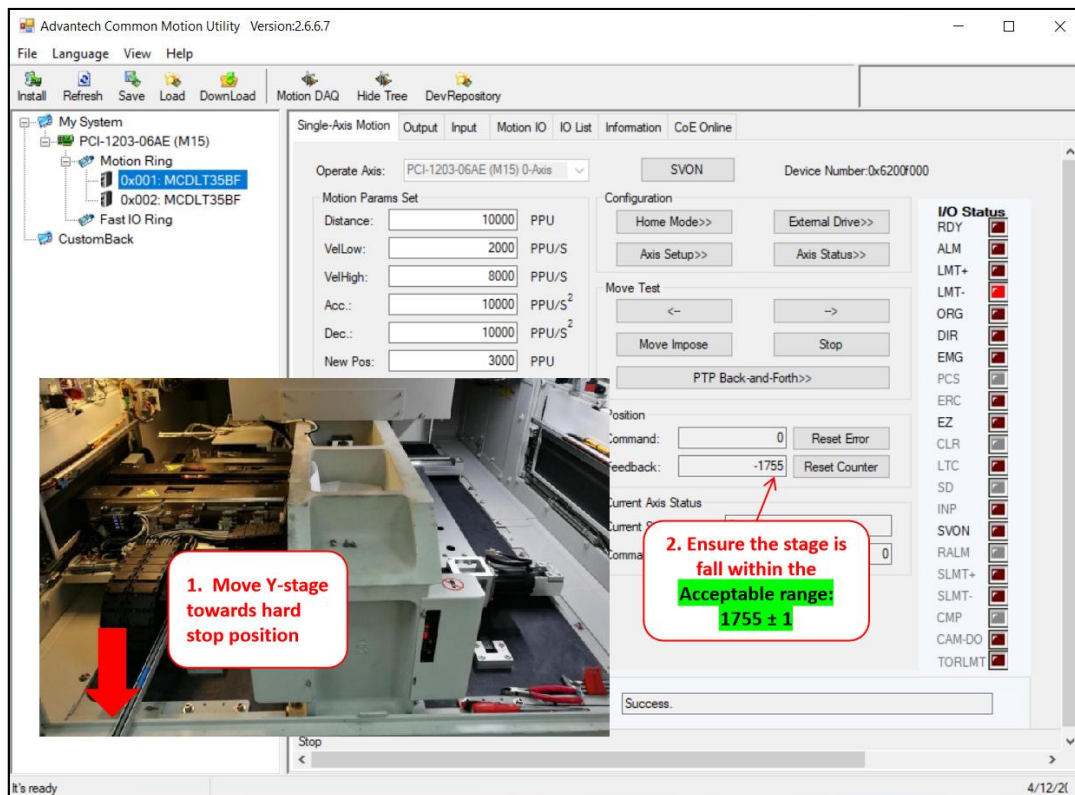


Figure 73: Move Y-Axis Stage to Feedback Position 1755 (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

81. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

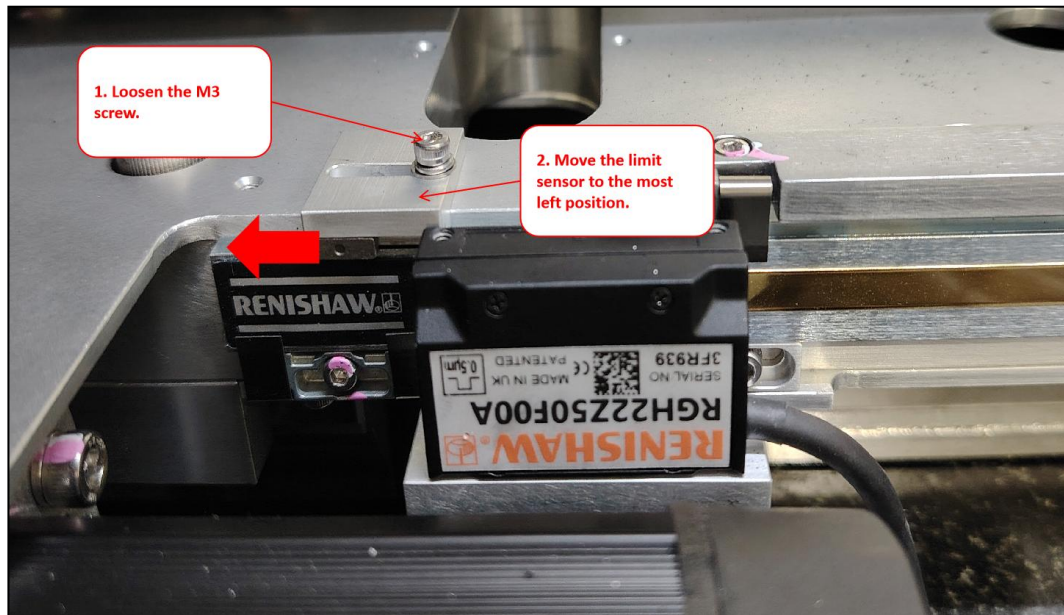


Figure 74: Move limit sensor to the Most Left Position (Sample Image)

82. **Adjust the limit sensor to right hand side until LMT- is triggered:** Slowly move the limit sensor towards the right hand side position until the LMT- signal is triggered.

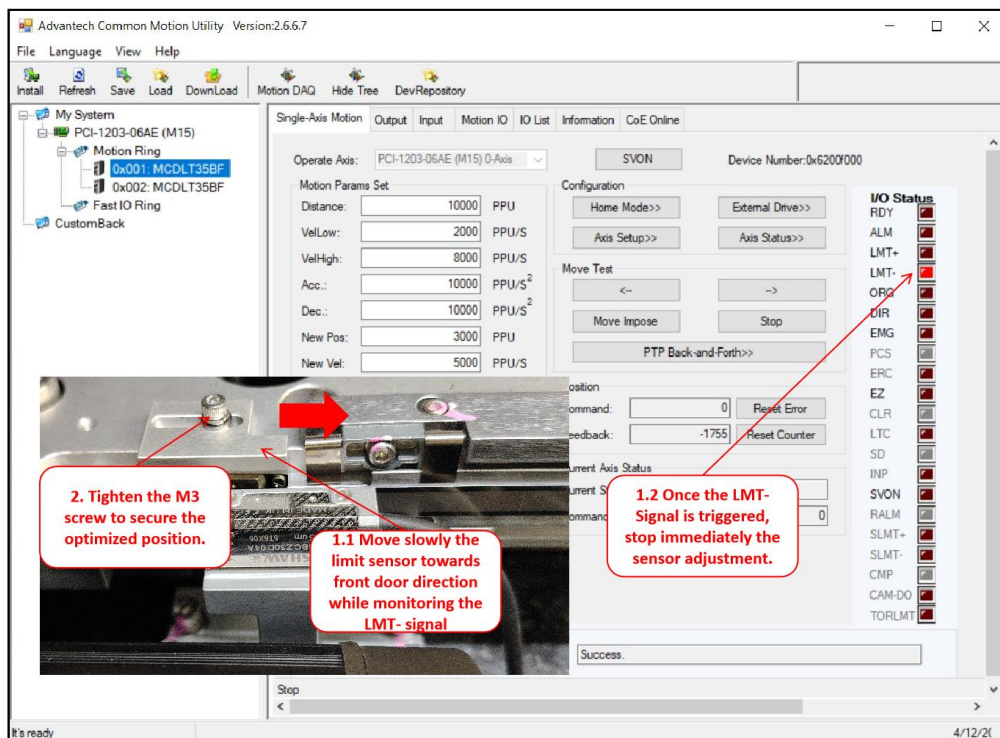


Figure 75: Positioning Limit Sensor to Triggered State (Sample Image)

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

83. **Proceed to Test Homing for Final Verification:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 4 for y-axis to test homing result. If any error, repeat all the required step all over again. Else, proceed to validate the current stage position.

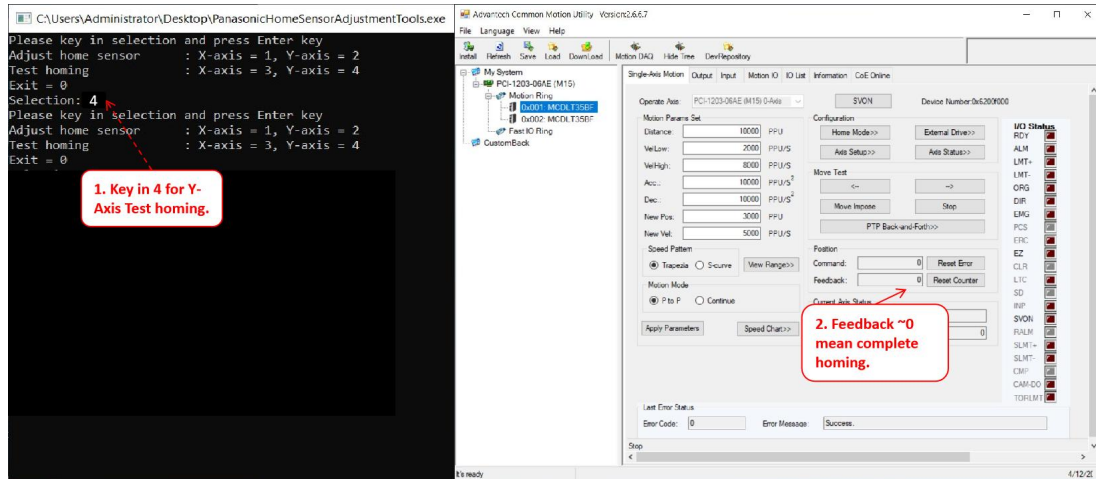


Figure 76: Perform Homing for Final Verification

84. **Re-connect the LAN cable labeled with Slice IO to the Atom PC:** Re-connect the LAN cable labeled with Slice IO to the Atom PC.



Figure 77: Reconnect the LAN Cable for Slice IO

85. **Go to Main Server and open v810:** Close the remote desktop connection and launch the v810 program.
86. **Panel Positioner Initialization:** Navigate to ‘Service > Subsystem Status & Control > Panel Positioner’. Click “Initialize”.

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

87. **Verify the “Current Stage Position” for Y-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the Y stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, route back to Step 64 to redo the adjustment all over again.

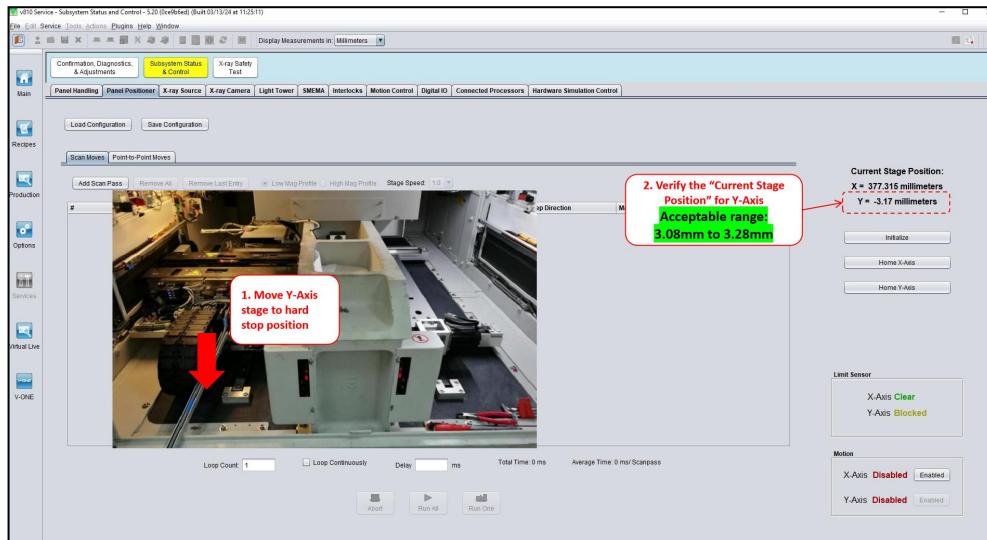


Figure 78: Current Stage Position for Y-Axis within range (Sample Image)

88. **Proceed to Validate Machine Performance:** Validation of machine if required before release the machine to production.

Note: If there is involvement of X-axis scale replacement, validation may be done upon all adjustments made.

	Work Instruction	Doc. Revision	01
	AXI V810 Quantic Home and Limit Sensors Alignment Procedure	Effective Date	30 Oct 2024

Validation of Machine Performance After Alignment for Sensors

89. **Tool Removal:** After completing the alignment process, ensure all tools, equipment, and debris are removed from the machine stage and surrounding area. Collect any tools used during the adjustment and confirmation procedures and return them to their designated storage locations.
90. **Access Door Switching:** Proceed to each access door of the machine and switch them back to production mode. Ensure that each access door is securely closed and locked according to standard operating procedures and verify that all safety interlocks associated with the access doors are engaged and functioning properly.
91. **Subsystem Initialization:** Power on the machine and run V810 software. Perform a login procedure to access the system interface. Navigate to the ‘Services > Subsystem Status & Control > Panel Positioner’ section within the software interface. Click on the “Initialize” button to reset the panel positioner subsystem to its default settings. Wait for the initialization process to complete before proceeding further.
92. **Long Scan Execution:** Execute five loops of long scan procedures using the machine's scanning capabilities. Ensure that the scan parameters are set according to the desired specifications and imaging requirements. Monitor the scan progress and observe the performance of the machine throughout the scanning process.
93. **CD&A Loop Execution:** Run one loop of CD&A (Comparison, Dosage, and Alignment) procedures to further validate the performance of the machine. Follow the standard operating procedures for CD&A loop execution, ensuring that all necessary parameters and settings are configured correctly.
94. **Monitoring and Verification of Machine Performance:** Monitor the execution of CD&A procedures and carefully observe the machine's performance, including accuracy, stability, and consistency in delivering results. Verify that the machine operates within acceptable tolerances and meets the required performance standards.